25th European Conference on Computational Biology (ECCB)

Tutorial session T13: 13:45 -17:30

Title: «Democratizing multi-omics data analysis with BiomiX and Nextflow: from single-omics to advanced integration»

Presenters :

Cristian IPERI PhD

USZ: Center for Experimental Rheumatology, Zurich Switzerland

Álvaro Fernández-Ochoa

University of Granada, Spain

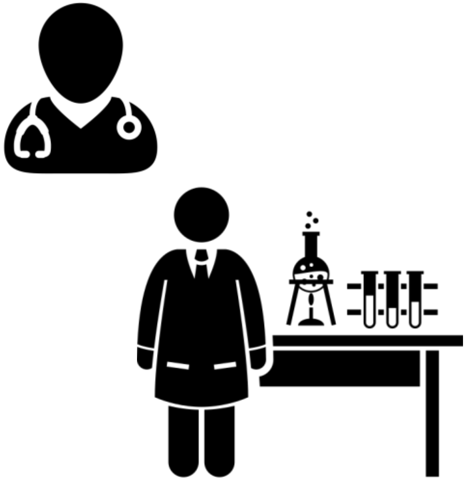
USZ Universitäts Spital Zürich

PRECISESADS
Molecular Reclassification to Find Clinically Useful Biomarkers for Systemic Autoimmune Diseases

Région BRETAGNE

LBAI U1227

03-09-2026



Access to data and methods

Clinicians, Specialists

- ✔ Deep knowledge of the disease
- ✘ Extract information from data

Bioinformaticians

- ✘ Deep knowledge of the disease
- ✔ Extract information from data

You have a biological question, but you are not a programmer

SpatialScope
Interactive Spatial Transcriptomics Analysis Platform

ShinyGO

NIH National Center for Advancing Translational Sciences

SEQUIN

MetaboAnalyst 6.0

Galaxy EUROPE

User-friendly tool for single omics analysis existing but Fragmented

Multiomics integration tools with accessible user interface

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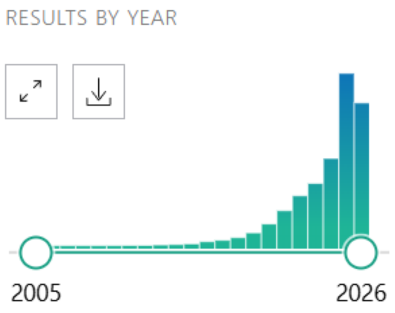


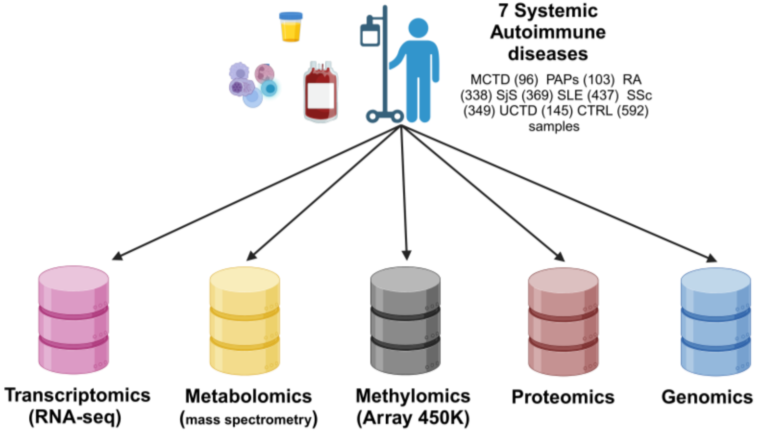
Multiomics research on pubmed
12,031 articles on 24 June 2026.

User-friendly tool for single omics analysis existing but Fragmented

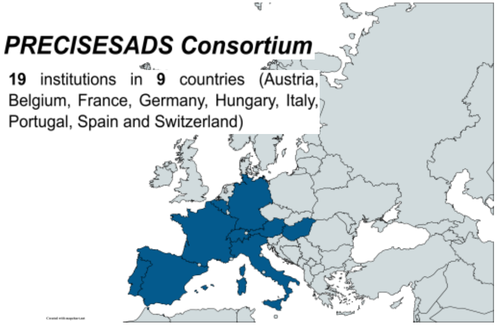
Multiomics integration tools with accessible user interface

PRECISESADS

Molecular Reclassification to Find Clinically Useful Biomarkers for Systemic Autoimmune Diseases

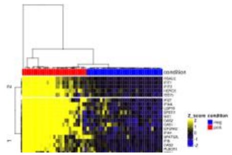


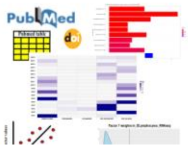
Rationale: Autoimmune disease clinical classifications show high heterogeneity within the disease.

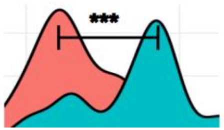


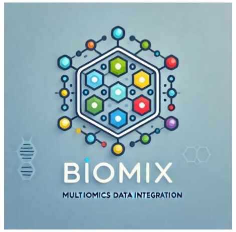
Objective: Reclassification of the autoimmunes diseases by biological and molecular criteria

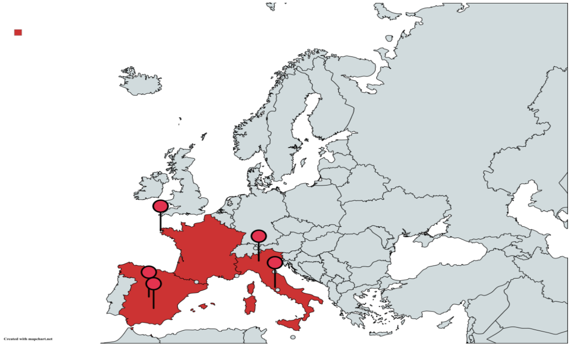
G. Barturen et al, Arthritis & Rheumatology (2021); L. Martin-Gutierrez, ACR (2021)











Software | Open access | Published: 10 January 2025

BiomiX, a user-friendly bioinformatic tool for democratized analysis and integration of multiomics data

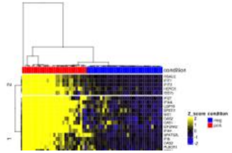
Cristian Iperi, Álvaro Fernández-Ochoa, Guillermo Barturen, Jacques-Olivier Pers, Nathan Foulquier, Eleonore Bettacchioli, Marta Alarcón-Riquelme, PRECISESADS Flow Cytometry Study Group, PRECISESADS Clinical Consortium, Divi Cornec, Anne Bordron & Christophe Jamin

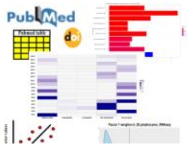
BMC Bioinformatics **26**, Article number: 8 (2025) | Cite this article

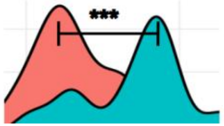
Website: https://ixi-97.github.io/

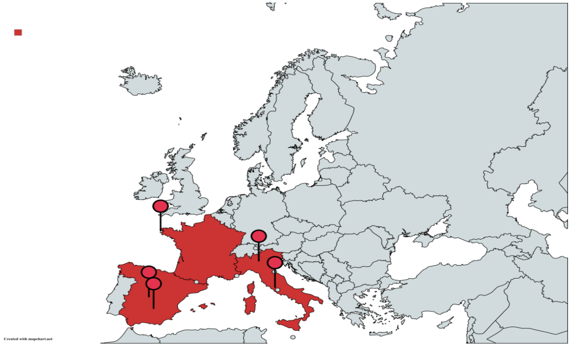
https://github.com/IxI-97/BiomiX

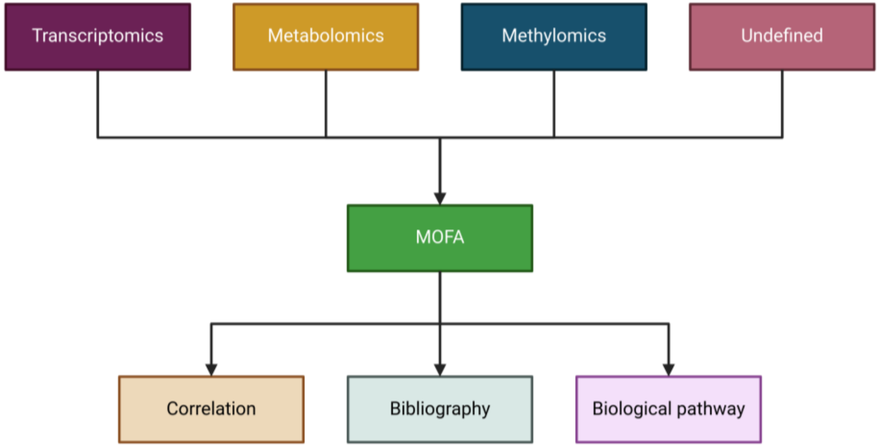
https://ixi-97.github.io/Tutorial.html











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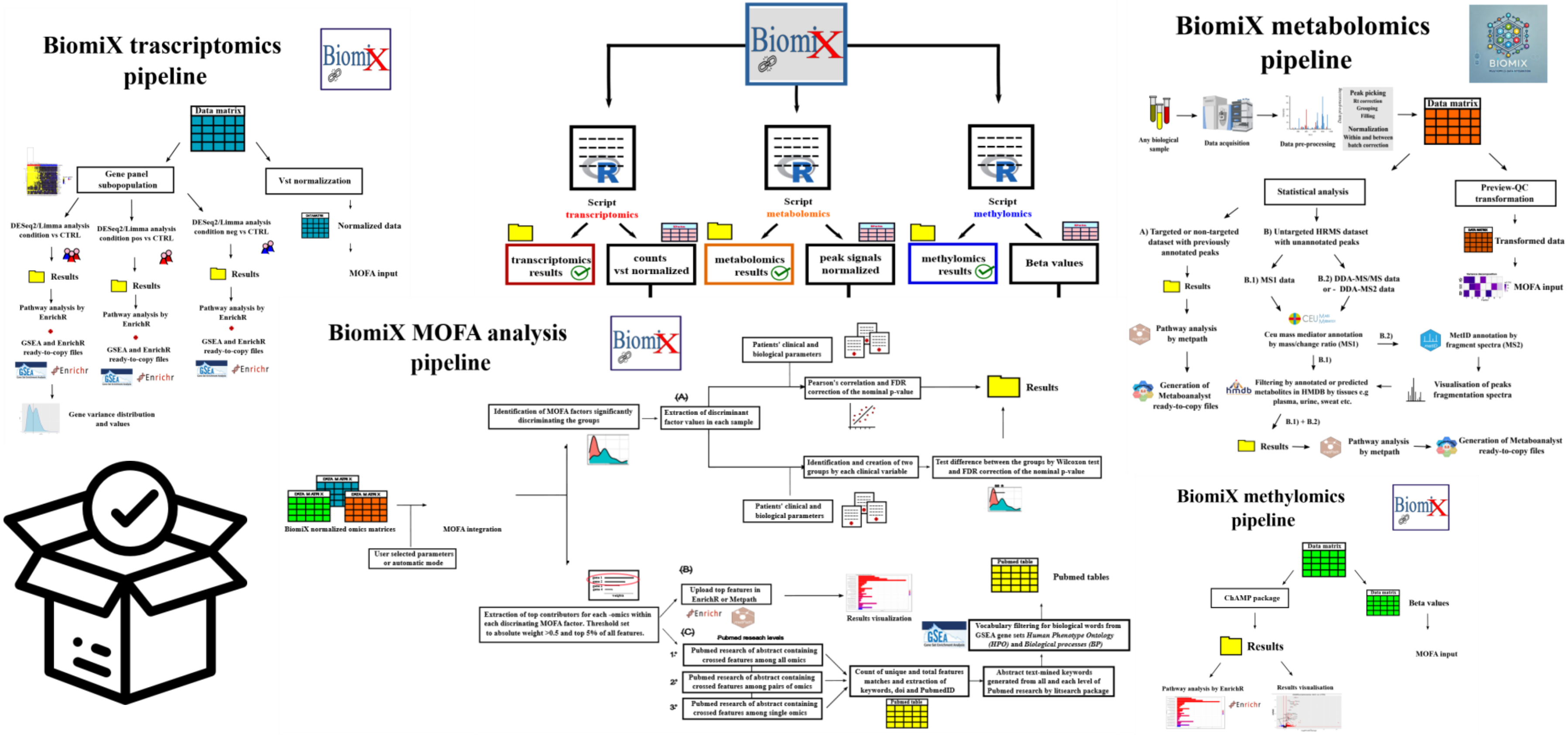
BiomiX trascriptomics pipeline
Data matrix
Gene panel subopopulation
Vst normalizzation
DESeq2/Limma analysis condition vs CTRL
DESeq2/Limma analysis condition pos vs CTRL
DESeq2/Limma analysis condition neg vs CTRL
Normalized data
MOFA input
Results
Pathway analysis by EnrichR
GSEA and EnrichR ready-to-copy files
Gene variance distribution and values
BiomiX
Script transcriptomics
Script metabolomics
Script methylomics
transcriptomics results
counts vst normalized
metabolomics results
peak signals normalized
methylomics results
Beta values
BiomiX MOFA analysis pipeline
Patients' clinical and biological parameters
Pearson's correlation and FDR correction of the nominal p-value
Results
Identification of MOFA factors significantly discriminating the groups
Extraction of discriminant factor values in each sample
Identification and creation of two groups by each clinical variable
Test difference between the groups by Wilcoxon test and FDR correction of the nominal p-value
BiomiX normalized omics matrices
MOFA integration
User selected parameters or automatic mode
Extraction of top contributors for each -omics within each discrinating MOFA factor. Threshold set to absolute weight >0.5 and top 5% of all features.
Upload top features in EnrichR or Metpath
Results visualization
Pubmed tables
Vocabulary filtering for biological words from GSEA gene sets Human Phenotype Ontology (HPO) and Biological processes (BP)
Pubmed research of abstract containing crossed features among all omics
Pubmed research of abstract containing crossed features among pairs of omics
Pubmed research of abstract containing crossed features among single omics
Count of unique and total features matches and extraction of keywords, doi and PubmedID
Abstract text-mined keywords generated from all and each level of Pubmed research by litsearch package
BiomiX metabolomics pipeline
Any biological sample
Data acquisition
Data pre-processing
Data matrix
Statistical analysis
Preview-QC transformation
A) Targeted or non-targeted dataset with previously annotated peaks
B) Untargeted HRMS dataset with unannotated peaks
Transformed data
MOFA input
B.1) MS1 data
B.2) DDA-MS/MS data or - DDA-MS2 data
Results
Pathway analysis by metpath
Ceu mass mediator annotation by mass/change ratio (MS1)
MetID annotation by fragment spectra (MS2)
Generation of Metaboanalyst ready-to-copy files
Filtering by annotated or predicted metabolites in HMDB by tissues e.g plasma, urine, sweat etc.
Visualisation of peaks fragmentation spectra
B.1) + B.2)
Results
Pathway analysis by metpath
Generation of Metaboanalyst ready-to-copy files
BiomiX methylomics pipeline
Data matrix
ChAMP package
Beta values
Results
MOFA input
Pathway analysis by EnrichR
Results visualisation

BiomiX, since then?

Hub of multiomics methods → **accessible to everyone**

Increase the community of developers and users

Look at the **single and spatial omics** revolution

BiomiX, since then?

LBAI U1227

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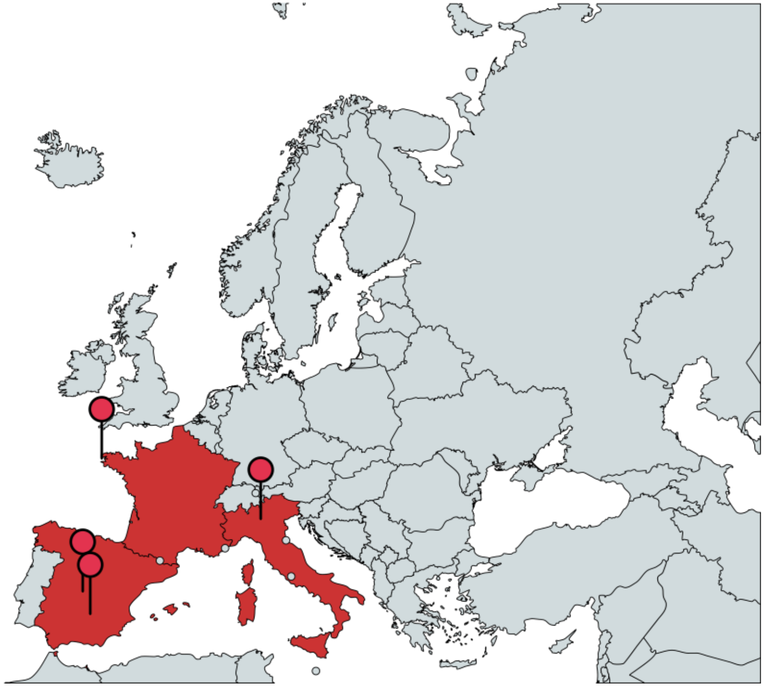
GGB UMR 1078

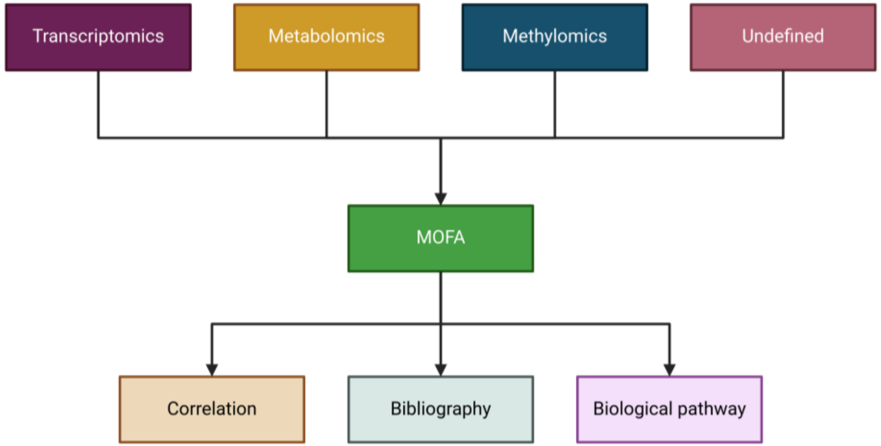
Université de Montréal

CEU Universidad San Pablo

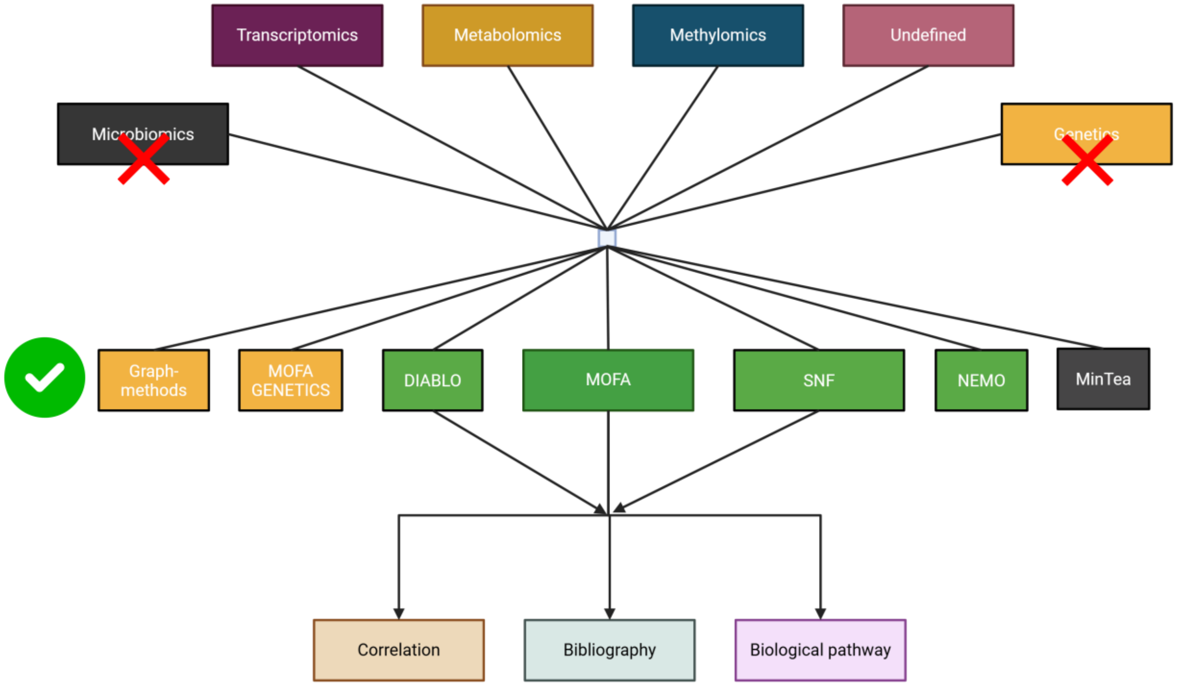
UNIVERSIDAD DE GRANADA



BiomiX consortium



BiomiX V2 2024



BiomiX 2026: Reality (V3.0)

Set up installation

BiomiX 3.0

Home Installation Tutorial Single-omics Integration

INSTALLATION

Get BiomiX 3.0 running

BiomiX 3.0 uses Docker to ensure a stable, reproducible environment across all platforms. The launcher scripts handle everything automatically, no command-line knowledge required.

0 Install Docker

BiomiX 3.0 requires Docker to be installed on your system. We recommend Docker Engine on Linux/Ubuntu, which is the environment BiomiX 3.0 has been built and tested for. New to Docker? Follow our Docker installation guide.

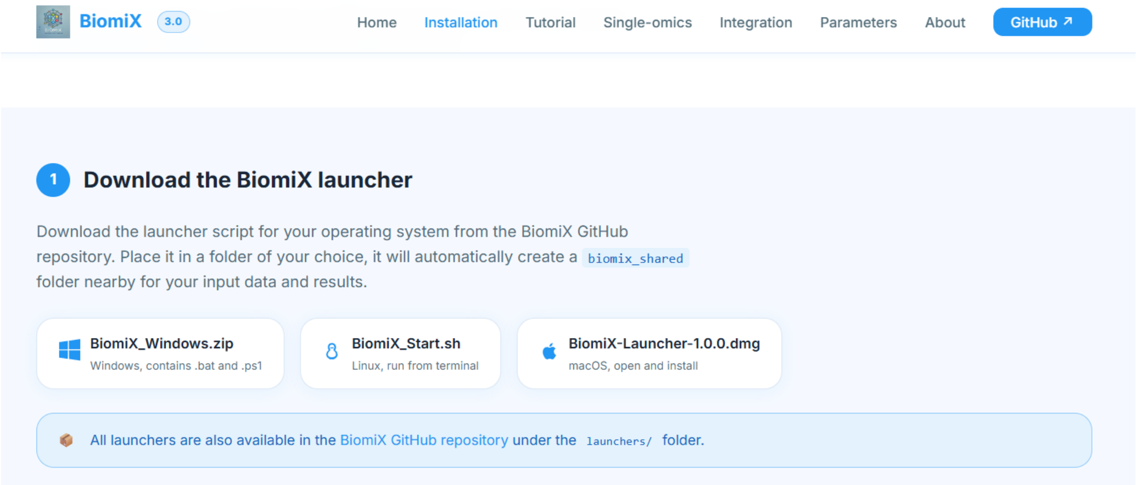
A local installation without Docker is still possible using the legacy install scripts. See the legacy installation section at the bottom of this page for details.

Slide to be updated with the final version*

1) Open BiomiX website:
https://biomix-consortium.github.io/

2) If not done yet install the Docker system for your system (Windows, MacOS, Linux)

Set up installation



Slide to be updated with the final version*

1) Open BiomiX website:
https://biomix-consortium.github.io/

2) If not done yet install the Docker system for your system (Windows, MacOS, Linux)

3) Download the BiomiX launcher

Set up installation

Slide to be updated with the final version*

2 Launch BiomiX

The launcher will automatically pull all required Docker modules on first run and open the BiomiX interface in your browser. Select your platform below.

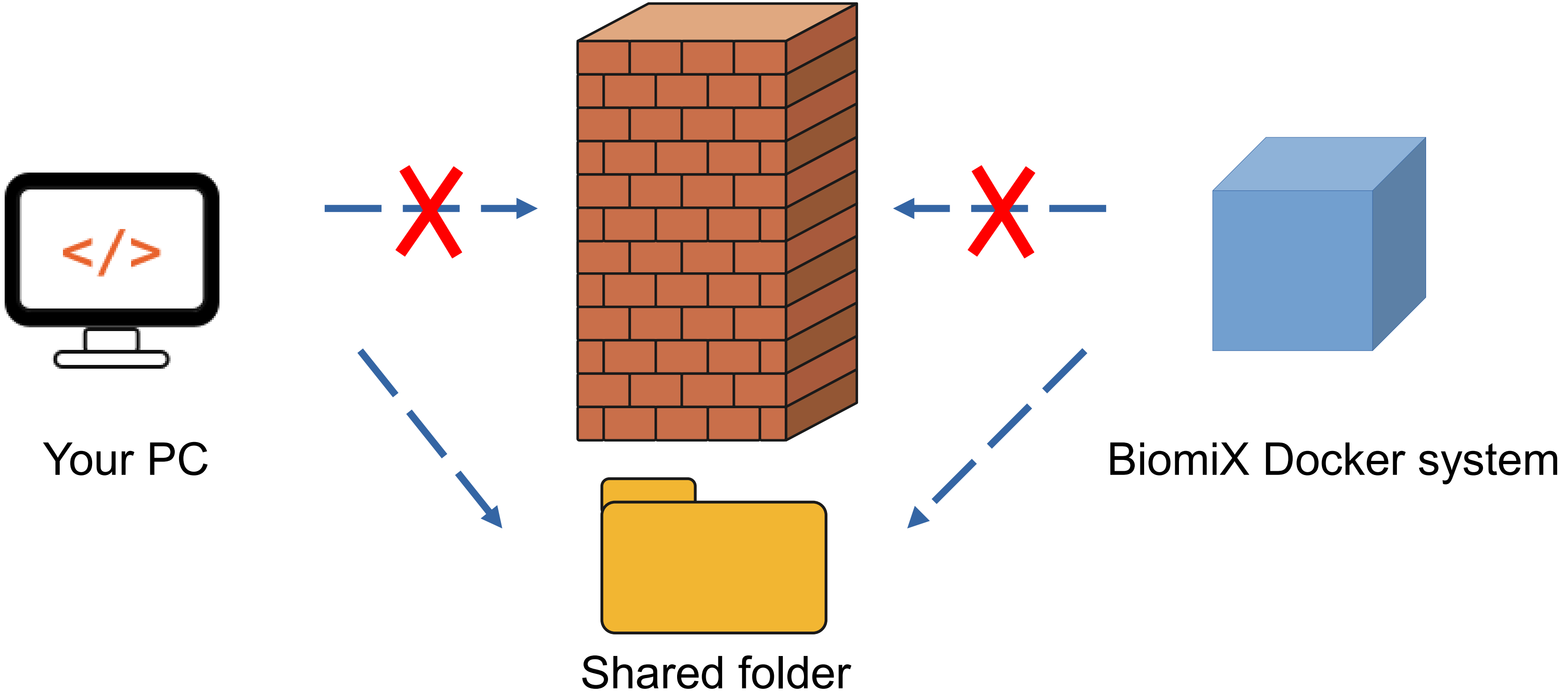
Windows | Linux | macOS

1. Make sure **Docker Desktop is open and running** (look for the whale icon in the system tray).
2. **Double-click** `BiomiX_Start.bat` , a setup window will appear.
3. Choose the **folder** where your input data is located and where results will be saved.
4. Optionally, enter your **NCBI API key** to enhance PubMed search in the biological interpretation module. This is not required, BiomiX works fully without it.
5. Click **Start**. BiomiX will open automatically in your browser at `http://localhost:3838` .

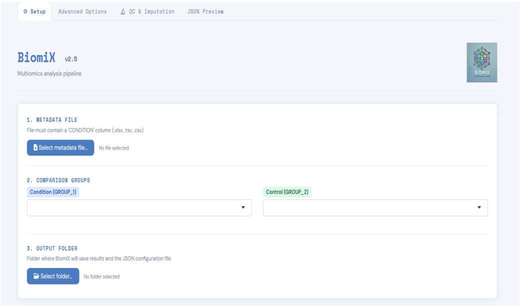
> 💡 Keep the terminal window open while using BiomiX. Closing it will stop the application.

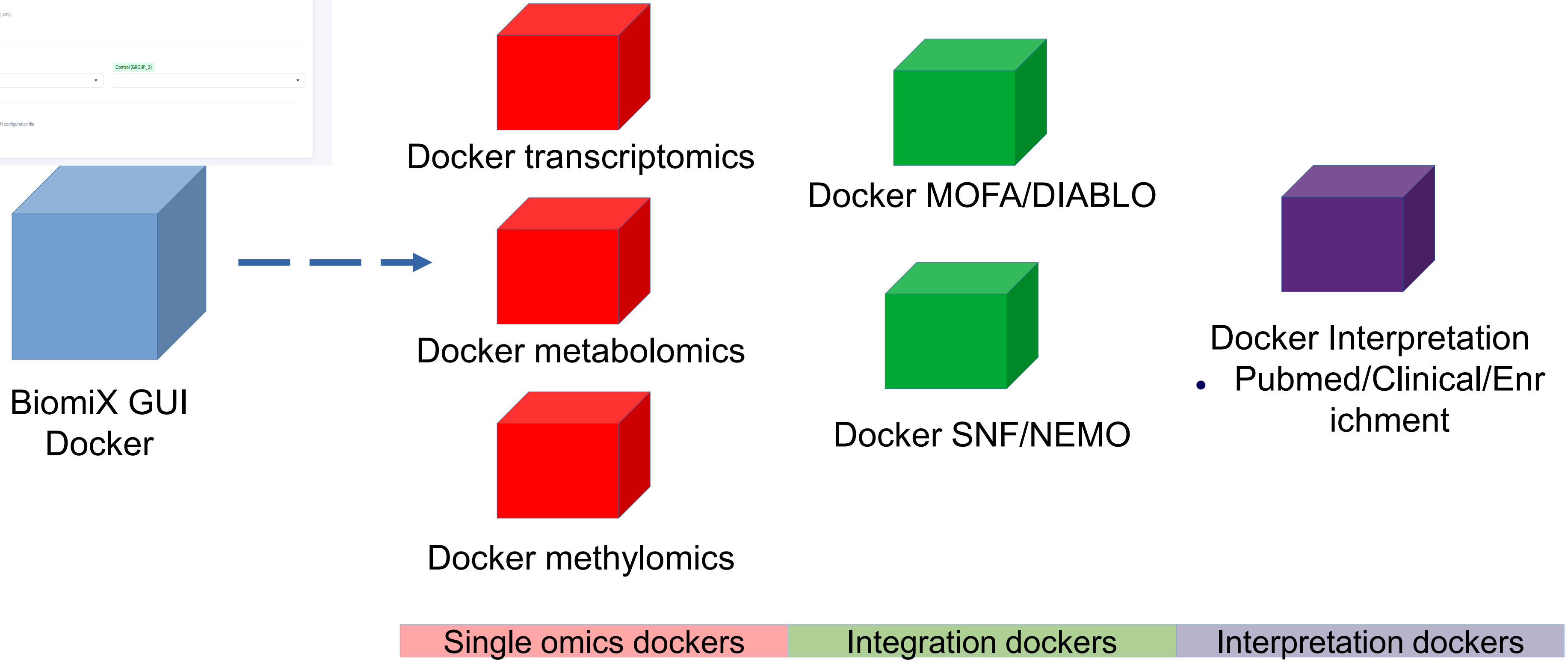
4) Launch BiomiX

How BiomiX works?

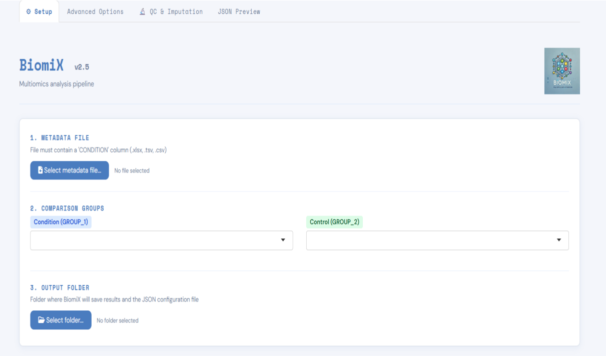


How BiomiX works?





How BiomiX works?



Json.file

Master script coordinating all the analysis.

BiomiX interface

Datasets of this tutorial

> J Clin Invest. 2018 Jan 2;128(1):427-445. doi: 10.1172/JCI93801. Epub 2017 Dec 11.

Drug-perturbation-based stratification of blood cancer

Sascha Dietrich [1 2 3 4], Małgorzata Oleś [1], Junyan Lu [1], Leopold Sellner [2 3], Simon Anders [1], Britta Velten [1], Bian Wu [3], Jennifer Hüllein [1 3], Michelle da Silva Liberio [3], Tatjana Walther [3], Lena Wagner [3], Sophie Rabe [1 2 3], Sonja Ghidelli-Disse [5], Marcus Bantscheff [5], Andrzej K Oleś [1], Mikołaj Słabicki [3], Andreas Mock [1], Christopher C Oakes [6 7], Shihui Wang [3], Sina Oppermann [3], Marina Lukas [3], Vladislav Kim [1], Martin Sill [8], Axel Benner [8], Anna Jauch [9], Lesley Ann Sutton [10], Emma Young [10], Richard Rosenquist [10], Xiyang Liu [3], Alexander Jethwa [3], Kwang Seok Lee [3], Joe Lewis [11], Kerstin Putzker [11], Christoph Lutz [2], Davide Rossi [12], Andriy Mokhir [13], Thomas Oellerich [14 15], Katja Zirlik [15 16], Marco Herling [17], Florence Nguyen-Khac [18], Christoph Plass [7 15], Emma Andersson [19], Satu Mustjoki [19], Christof von Kalle [3 15 20], Anthony D Ho [2], Manfred Hensel [21], Jan Dürig [15 22], Ingo Ringshausen [23], Marc Zapatka [24], Wolfgang Huber [1 4], Thorsten Zenz [2 3 15 25]

Affiliations + expand

PMID: 29227286 PMCID: PMC5749541 DOI: 10.1172/JCI93801

> Microbiol Spectr. 2023 Aug 17;11(4):e0057723. doi: 10.1128/spectrum.00577-23. Epub 2023 Jul 31.

Integration of Metabolomics and Transcriptomics Reveals Major Metabolic Pathways and Potential Biomarkers Involved in Pulmonary Tuberculosis and Pulmonary Tuberculosis-Complicated Diabetes

Yunguang Wang [1], Xinxin He [2], Danna Zheng [3], Qiang He [4], Lifang Sun [5 6], Juan Jin [2 4]

Affiliations + expand

PMID: 37522815 PMCID: PMC10434036 DOI: 10.1128/spectrum.00577-23

98 CLL IGHV mutated

74 CLL IGHV unmutated

? 12 CLL IGHV unknown

13 HC

13 PTB

13 PTB and DM

1) Chronic lymphocytic leukemia (CLL): EGAS00001001746

2) Pulmonary tubercolosis and diabetes complications: MTBLS7623

Data generation

Data alignment/QC

Gene count

DGE analysis

Enrichment analysis

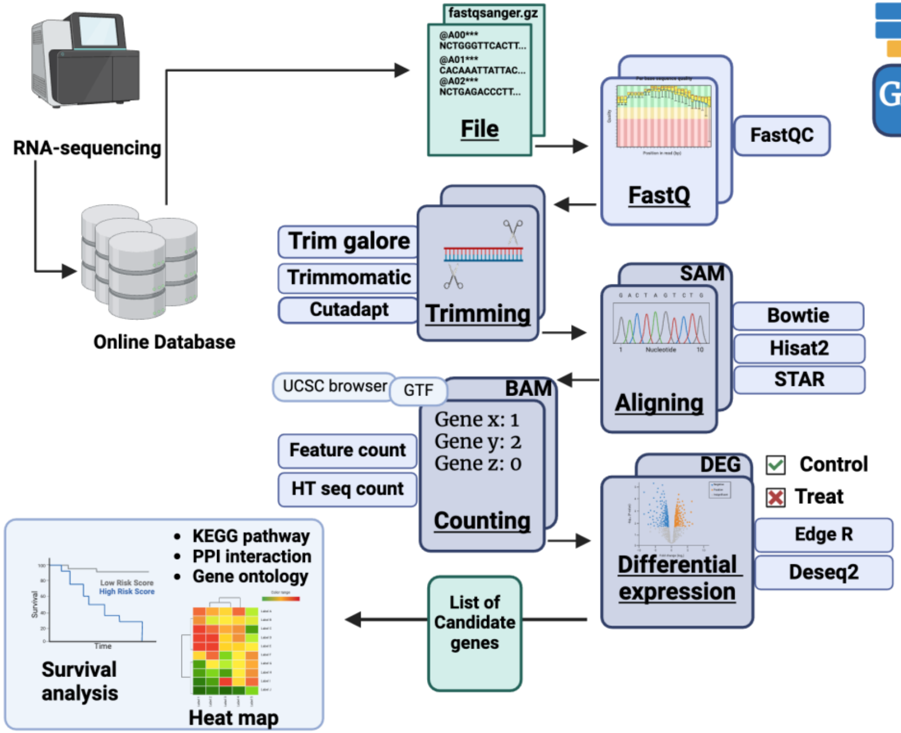
slurm
workload manager

Not Supported by BiomiX
Cluster/High performace
computer running

BIOMIX
MULTIOMICS DATA INTEGRATION

BiomiX Supported!
Locally running

Transcriptomics analysis CLL

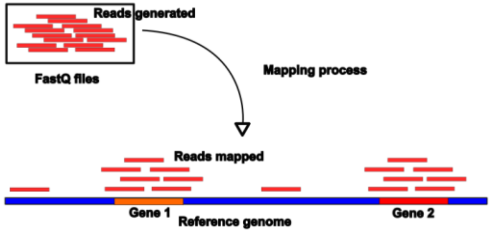


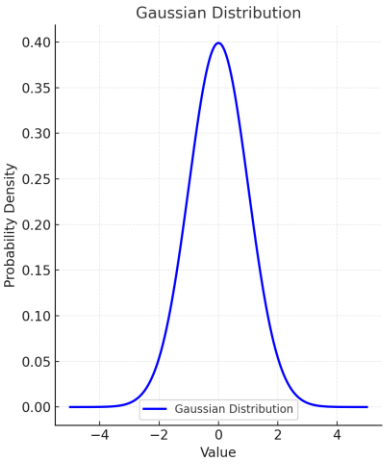
GALAXY.EU

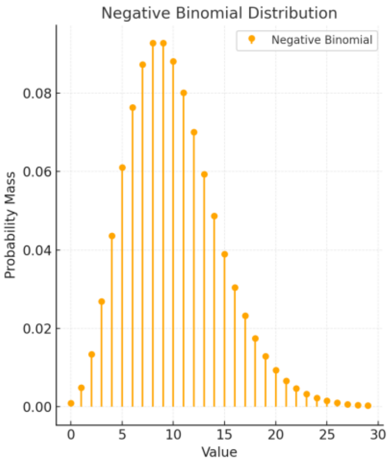
https://usegalaxy.eu/

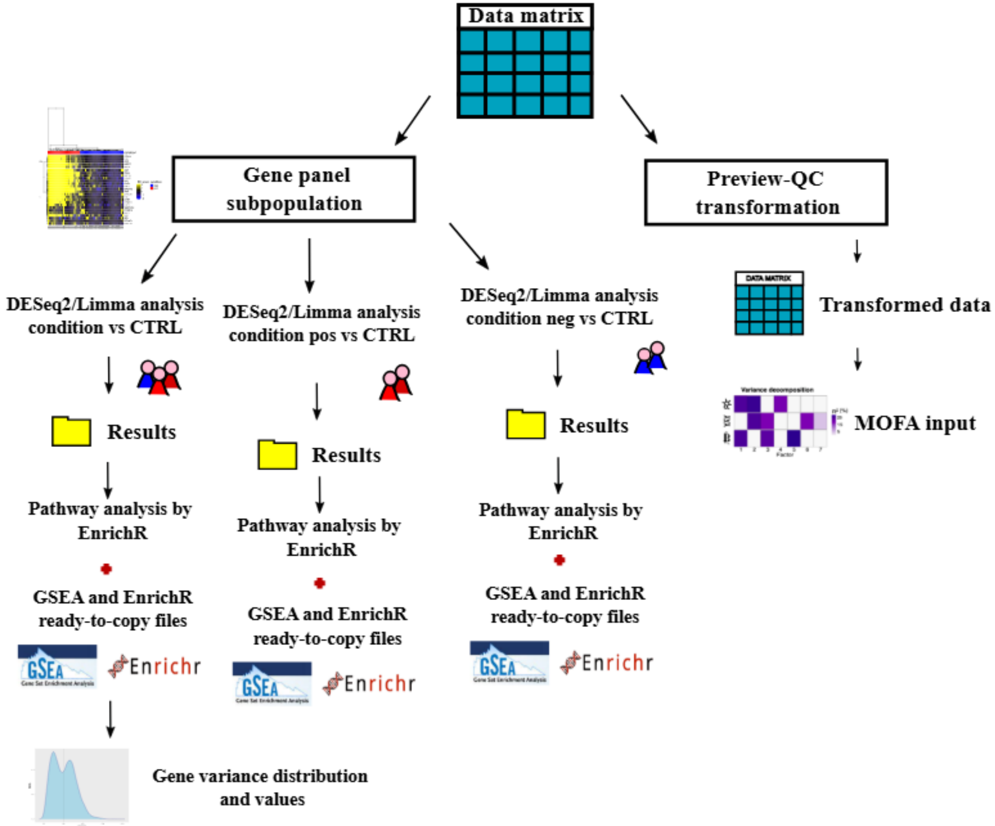
Image adapted from: Sabbaghian A. (2024). User friendly procedure for conducting bulk RNA sequencing analysis using the GALAXY.eu platform. Biorender

BiomiX transcriptomics pipeline







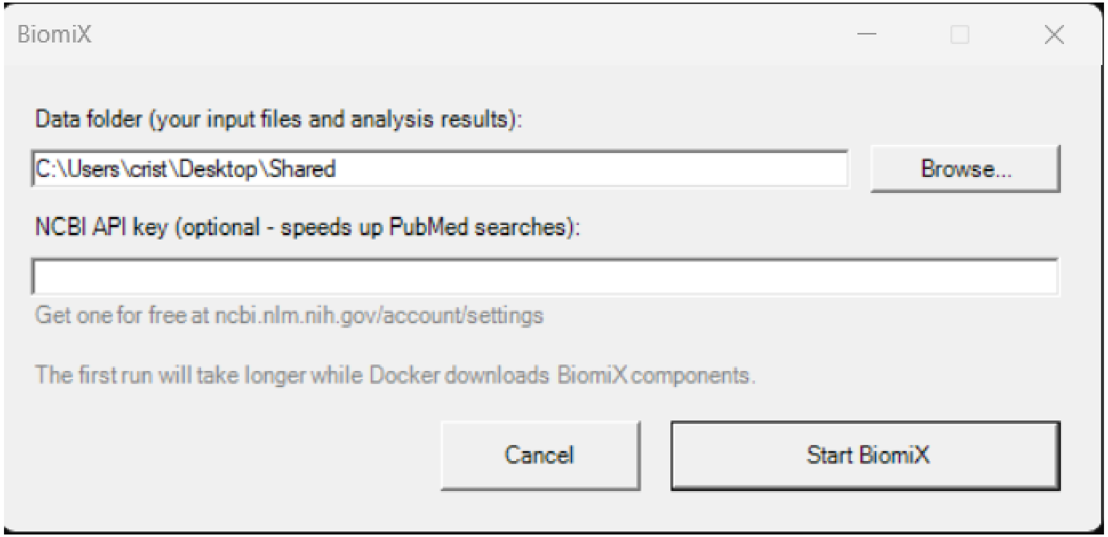


Transcriptomics tutorial

https://sagarsajeev.medium.com/business-logic-vulnerability-via-idor-6d510f1caea9

Transcriptomics analysis CLL

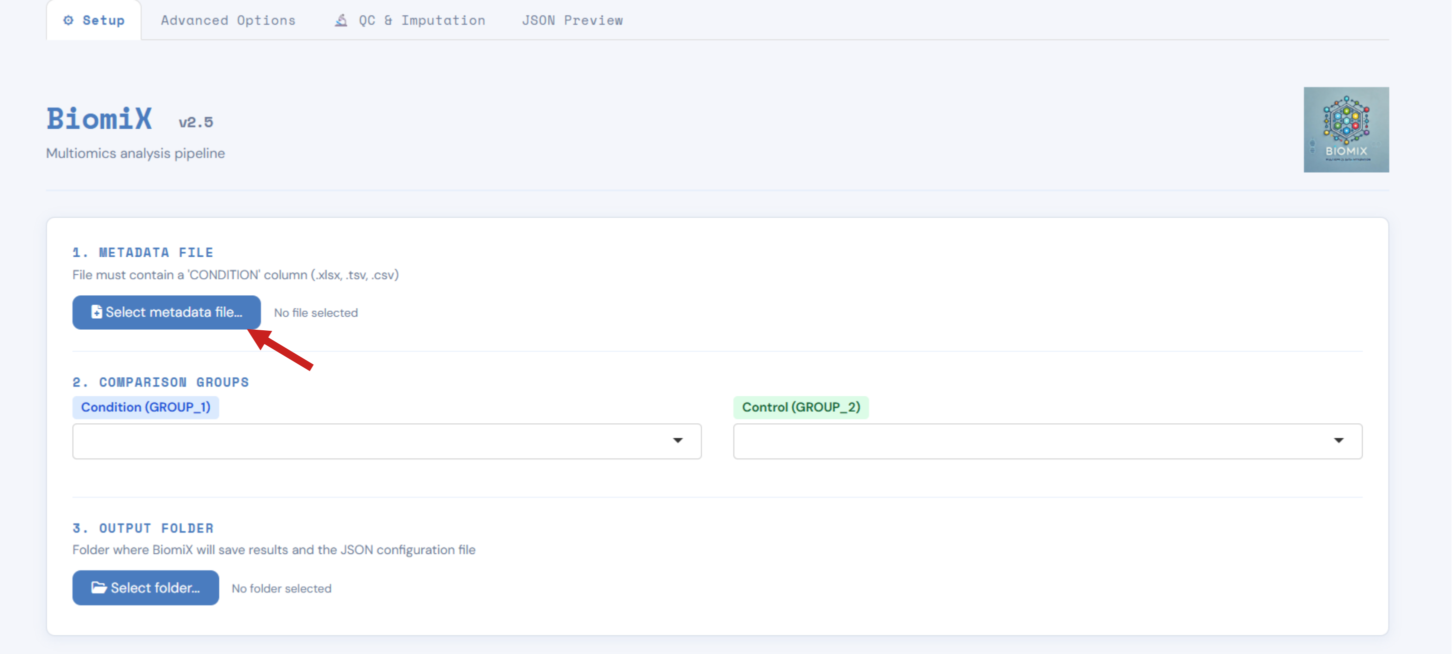
1) Open
BiomiX_Start.bat (Windows) **or**
BiomiX_Start.sh (Linux) **or**
BiomiX-Launcher-1.0.0.dmg (MacOS)



- Create a folder in Desktop called "**shared**" and set it as BiomiX reference directory.

Transcriptomics analysis CLL

1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder.



Transcriptomics analysis CLL

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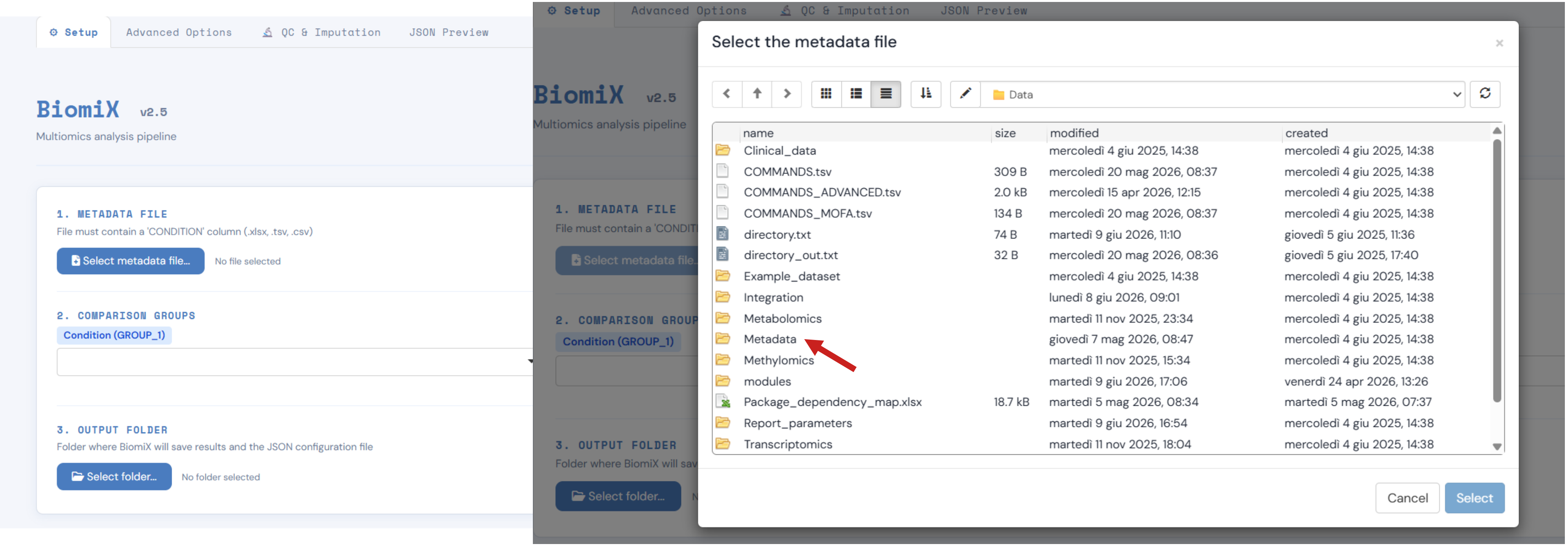
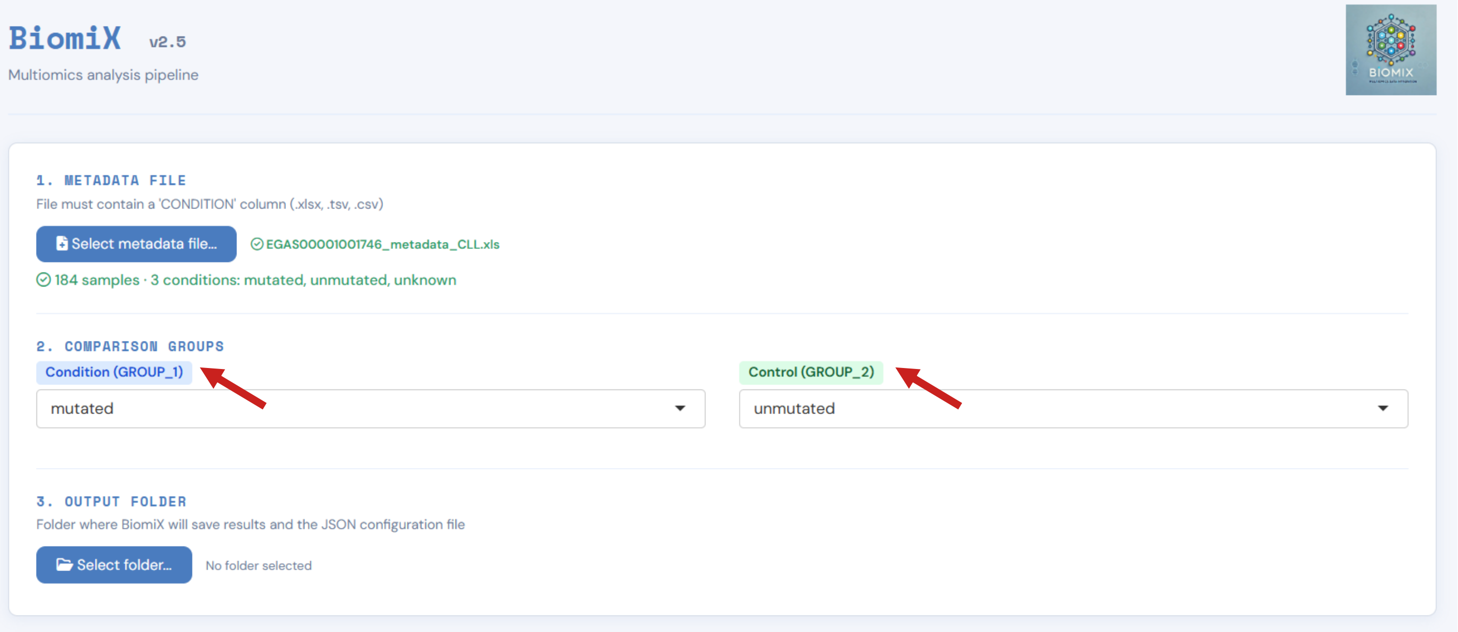


Transcriptomics analysis CLL

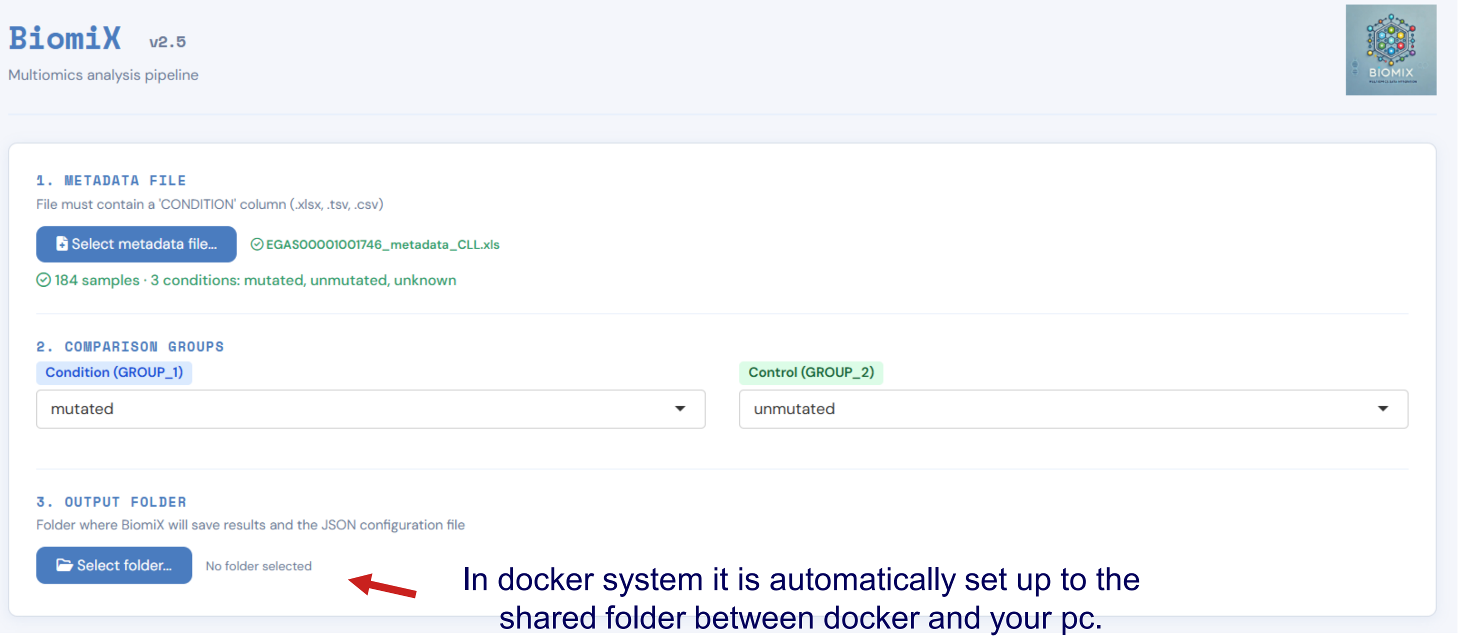
1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder.



BiomiX need two metadata columns:
"**ID**": Unique sample ID
"**CONDITION**": Sample condition

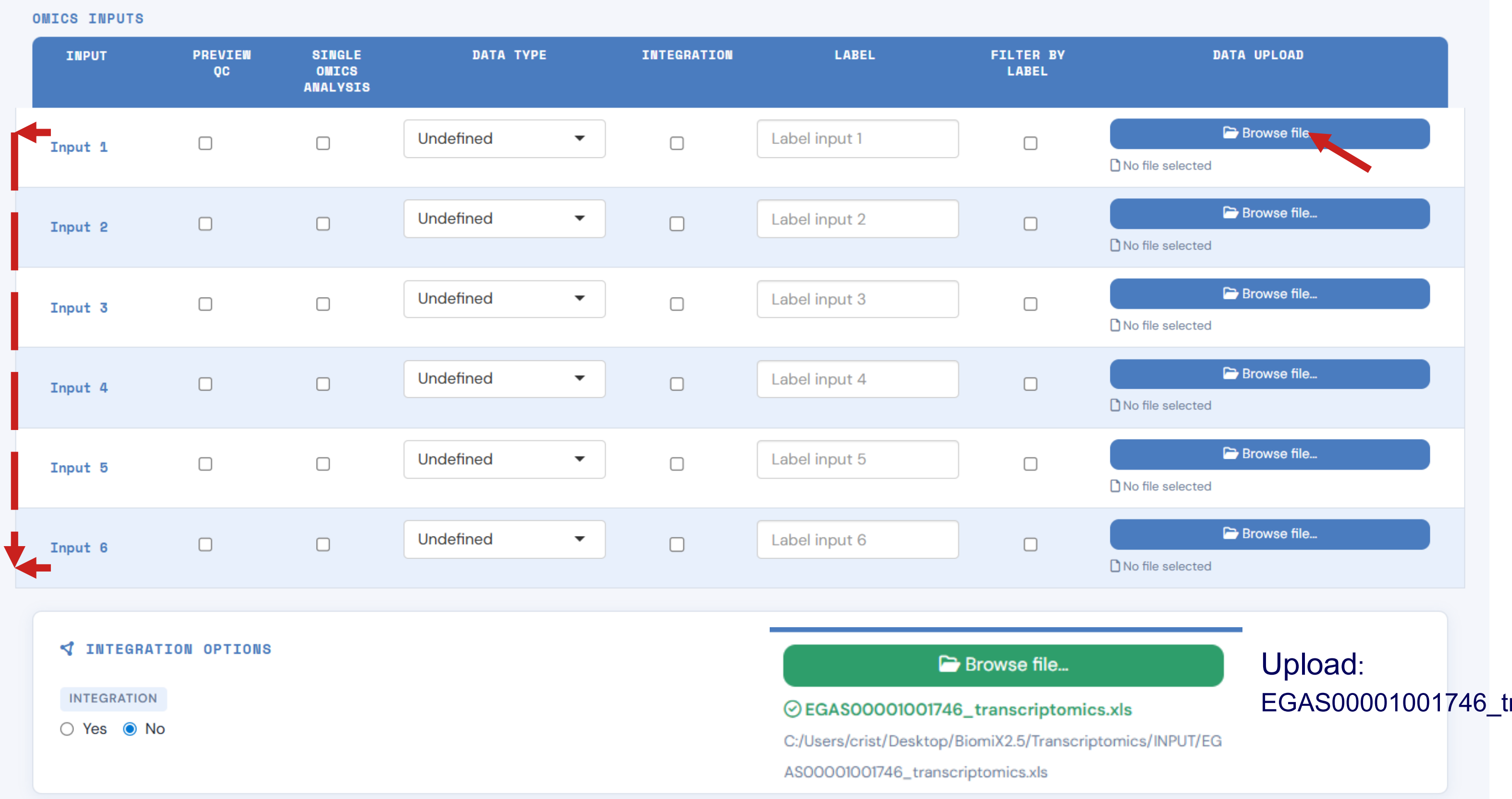
Transcriptomics analysis CLL

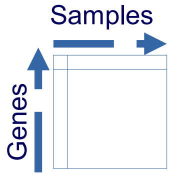
1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder.



In docker system it is automatically set up to the shared folder between docker and your pc.

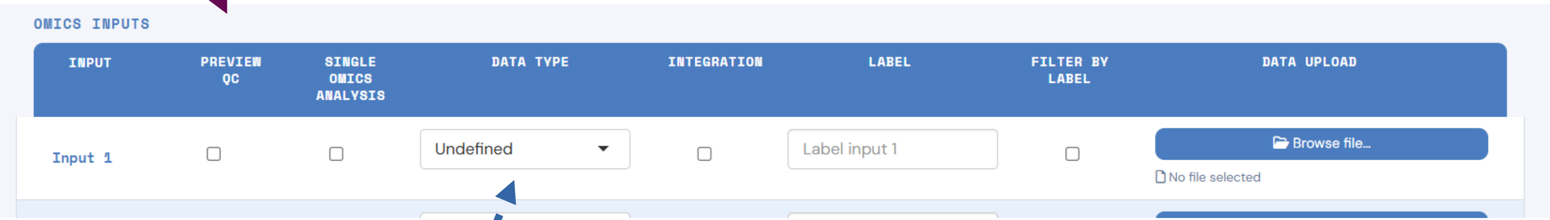
Slot system:
6 slot





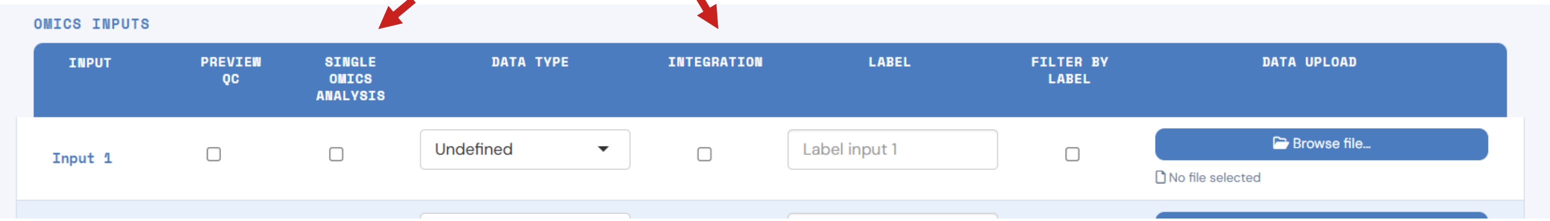
Upload:
EGAS00001001746_transcriptomics.xls

Normalization and visualisation of the data before the analysis (Shiny app).

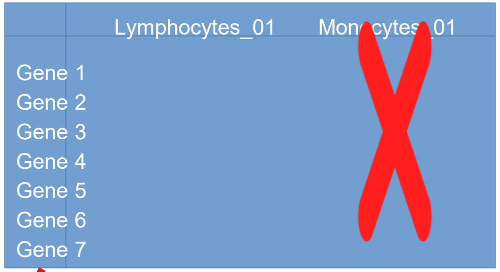


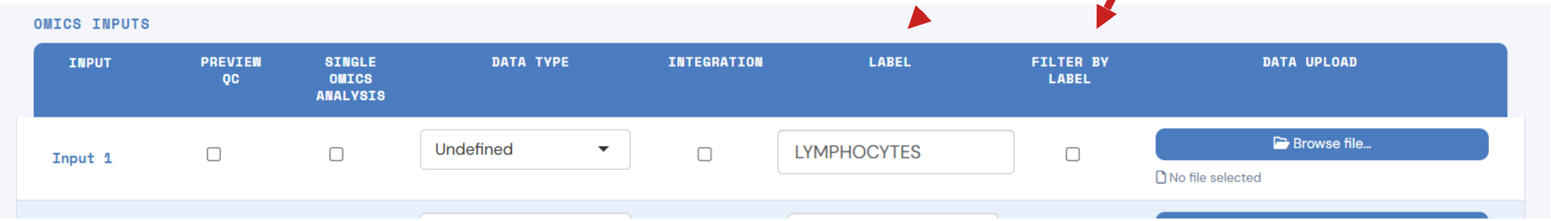
Which omics? If not included select undefined (default)

Do you want to perform a single omics analysis or use the slot for the integration?

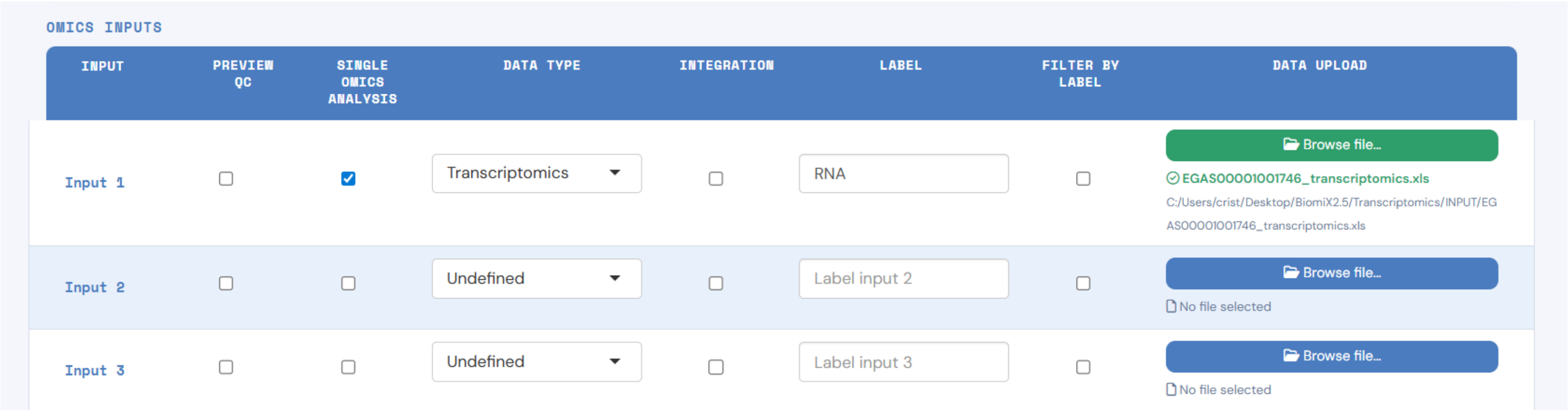


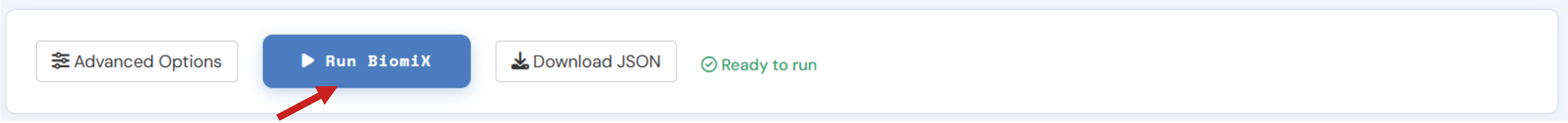
Personalized name for your dataset. If “filter” selected will only keep the sample sames containing this label.



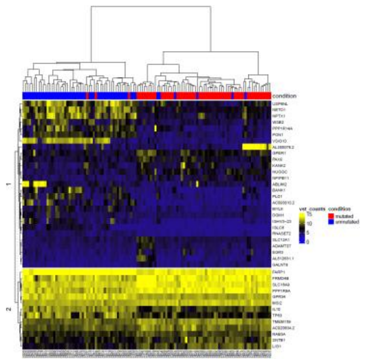


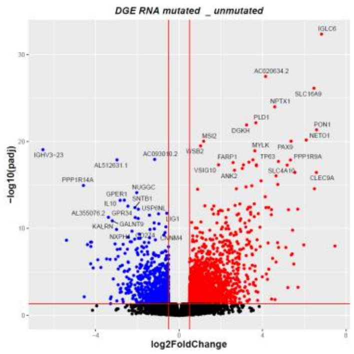
Run your first analysis





Check in Transcriptomics/Output folder → RNA_mutated_vs_unmutated created!





Pathway_analysis	22/01/2026 11:10	Cartella di file	
Subpopulation_input	22/01/2026 11:09	Cartella di file	
Heatmap_top_genes_RNA_unmutated_vs...	28/04/2026 19:49	Adobe Acrobat D...	41 KB
mutated_unmutated_RNA.tsv	28/04/2026 19:48	File TSV	5.643 KB
mutated_unmutated_RNA_RAW.gct	28/04/2026 19:49	File GCT	58.409 KB
mutated_unmutated_RNA_RAW_meta.tsv	28/04/2026 19:49	File TSV	4 KB
mutated_unmutated_RNA_RAW_meta_gr...	28/04/2026 19:49	File TSV	2 KB
plot_DGE_mutated_unmutated_RNA.pdf	28/04/2026 19:49	Adobe Acrobat D...	1.597 KB
RNA_mutated_unmutated_DOWN_ENRIC...	28/04/2026 19:49	File TSV	8 KB
RNA_mutated_unmutated_GENESDOWN....	28/04/2026 19:49	File TSV	115 KB
RNA_mutated_unmutated_GENESUP.tsv	28/04/2026 19:49	File TSV	274 KB
RNA_mutated_unmutated_UP_ENRICHR.tsv	28/04/2026 19:49	File TSV	19 KB

Input for GSEA program

DGE results all

	baseMean	log2FoldCh	lfcSE	stat	pvalue	padj
ENSG0000	1906.988	6.828504	0.533117	12.80864	1.4667914	4.56362816482032e-33
ENSG0000	6.345746	4.146017	0.349858	11.85056	2.1377006	3.32551409152136e-28
ENSG0000	111.8656	6.465554	0.559783	11.55012	7.3721385	7.64564487909312e-27
ENSG0000	386.3032	4.586128	0.413405	11.09356	1.3481534	1.04862742655686e-24

Pathway_analysis	22/01/2026 11:10	Cartella di file	
Subpopulation_input	22/01/2026 11:09	Cartella di file	
Heatmap_top_genes_RNA_unmutated_vs...	28/04/2026 19:49	Adobe Acrobat D...	41 KB
mutated_unmutated_RNA.tsv	28/04/2026 19:48	File TSV	5.643 KB
mutated_unmutated_RNA_RAW.gct	28/04/2026 19:49	File GCT	58.409 KB
mutated_unmutated_RNA_RAW_meta.tsv	28/04/2026 19:49	File TSV	4 KB
mutated_unmutated_RNA_RAW_meta_gr...	28/04/2026 19:49	File TSV	2 KB
plot_DGE_mutated_unmutated_RNA.pdf	28/04/2026 19:49	Adobe Acrobat D...	1.597 KB
RNA_mutated_unmutated_DOWN_ENRIC...	28/04/2026 19:49	File TSV	8 KB
RNA_mutated_unmutated_GENESDOWN....	28/04/2026 19:49	File TSV	115 KB
RNA_mutated_unmutated_GENESUP.tsv	28/04/2026 19:49	File TSV	274 KB
RNA_mutated_unmutated_UP_ENRICHR.tsv	28/04/2026 19:49	File TSV	19 KB

Only negative

Only Positive

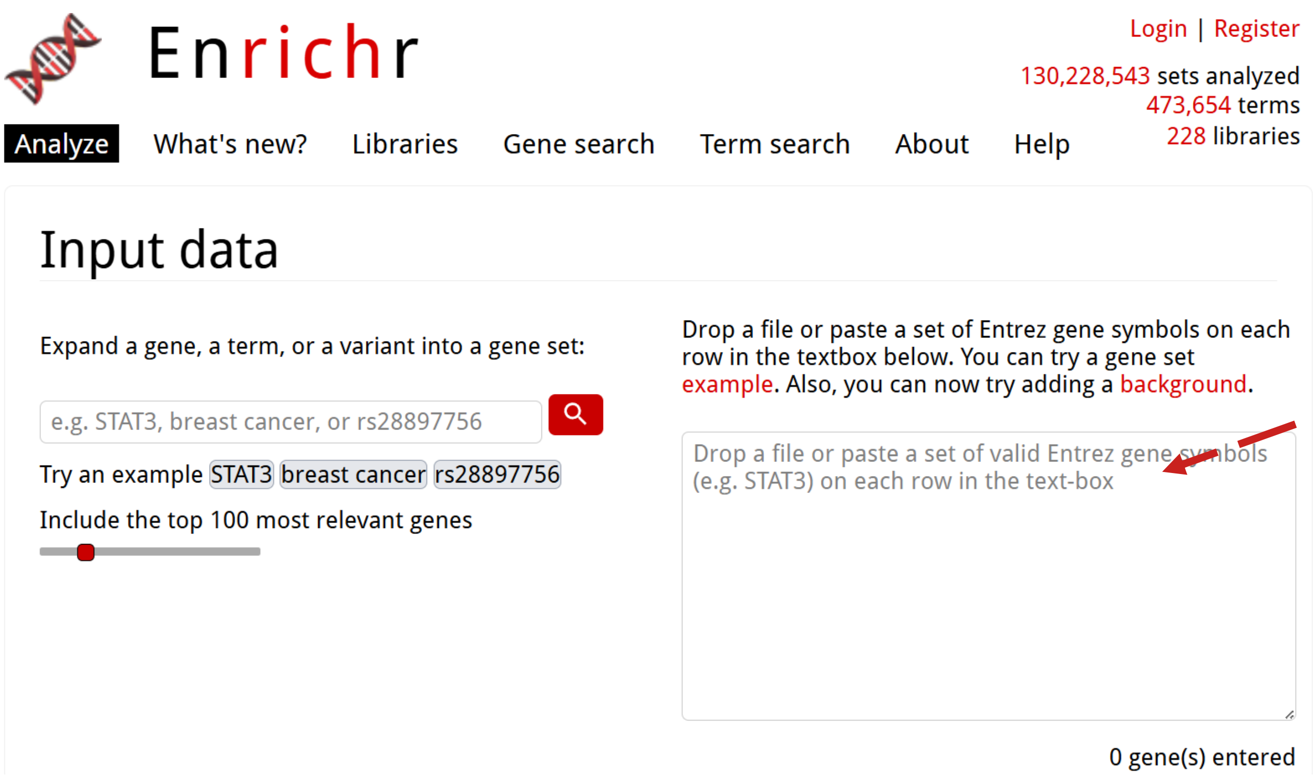
Input for GSEA program

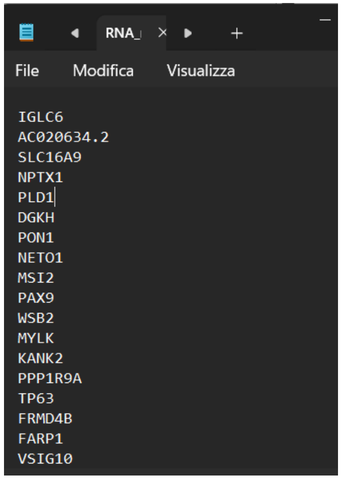
GSEA input

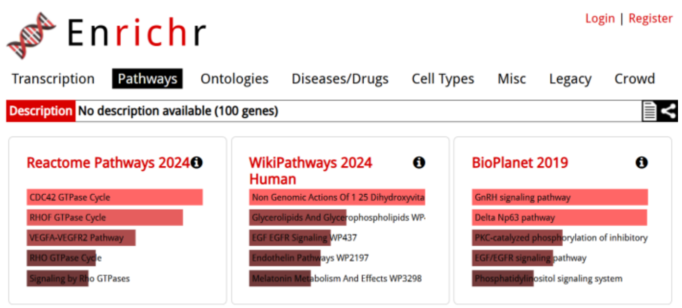
Input negative gene EnrichR

Input positive gene EnrichR

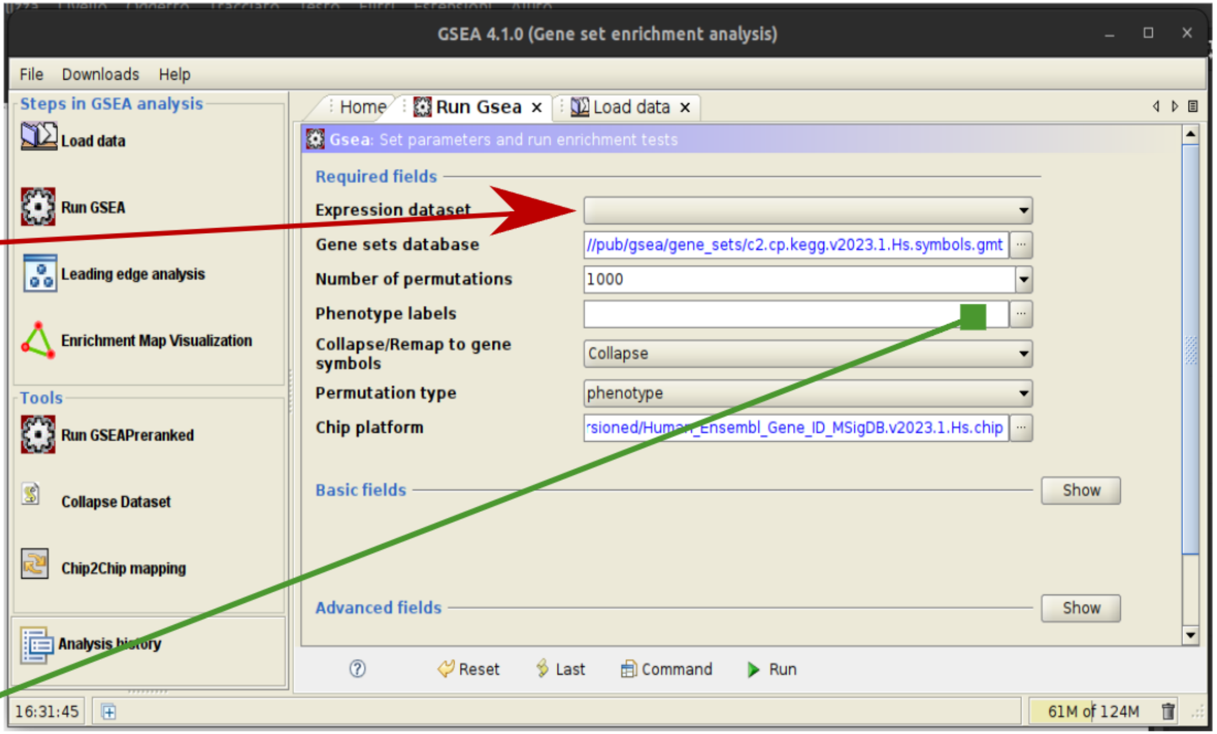
Pathway_analysis	22/01/2026 11:10	Cartella di file	
Subpopulation_input	22/01/2026 11:09	Cartella di file	
Heatmap_top_genes_RNA_unmutated_vs...	28/04/2026 19:49	Adobe Acrobat D...	41 KB
mutated_unmutated_RNA.tsv	28/04/2026 19:48	File TSV	5.643 KB
mutated_unmutated_RNA_RAW.gct	28/04/2026 19:49	File GCT	58.409 KB
mutated_unmutated_RNA_RAW_meta.tsv	28/04/2026 19:49	File TSV	4 KB
mutated_unmutated_RNA_RAW_meta_gr...	28/04/2026 19:49	File TSV	2 KB
plot_DGE_mutated_unmutated_RNA.pdf	28/04/2026 19:49	Adobe Acrobat D...	1.597 KB
RNA_mutated_unmutated_DOWN_ENRIC...	28/04/2026 19:49	File TSV	8 KB
RNA_mutated_unmutated_GENESDOWN....	28/04/2026 19:49	File TSV	115 KB
RNA_mutated_unmutated_GENESUP.tsv	28/04/2026 19:49	File TSV	274 KB
RNA_mutated_unmutated_UP_ENRICHR.tsv	28/04/2026 19:49	File TSV	19 KB


Enrichr
Login | Register
130,228,543 sets analyzed
473,654 terms
228 libraries
Analyze
What's new?
Libraries
Gene search
Term search
About
Help
Input data
Expand a gene, a term, or a variant into a gene set:
e.g. STAT3, breast cancer, or rs28897756
Try an example STAT3 breast cancer rs28897756
Include the top 100 most relevant genes
Drop a file or paste a set of Entrez gene symbols on each row in the textbox below. You can try a gene set example. Also, you can now try adding a background.
Drop a file or paste a set of valid Entrez gene symbols (e.g. STAT3) on each row in the text-box
0 gene(s) entered


RNA_
File
Modifica
Visualizza
IGLC6
AC020634.2
SLC16A9
NPTX1
PLD1
DGKH
PON1
NETO1
MSI2
PAX9
WSB2
MYLK
KANK2
PPP1R9A
TP63
FRMD4B
FARP1
VSIG10


Enrichr
Login | Register
Transcription
Pathways
Ontologies
Diseases/Drugs
Cell Types
Misc
Legacy
Crowd
Description No description available (100 genes)
Reactome Pathways 2024
CDC42 GTPase Cycle
RHOF GTPase Cycle
VEGFA-VEGFR2 Pathway
RHO GTPase Cycle
Signaling by Rho GTPases
WikiPathways 2024 Human
Glycerolipids And Glycerophospholipids WP-
EGF EGFR Signaling WP437
Endothelin Pathways WP2197
Melatonin Metabolism And Effects WP3298
BioPlanet 2019
GnRH signaling pathway
Delta Np63 pathway
PKC-catalyzed phosphorylation of inhibitory
EGF/EGFR signaling pathway
Phosphatidylinositol signaling system

	A	B	C	D	E
1	#1.2				
2	52141	28			
3	NAME	Description	32140242	32140244	32141679
4	ENSG00000000003	NA	0	1.03138546799503	5.25875398402679
5	ENSG00000000005	NA	0	0	3.9440654880201
6	ENSG00000000419	NA	224.658295044468	207.308479067001	215.608913345099
7	ENSG00000000457	NA	411.72436542014	319.729495078459	337.874943473722
8	ENSG00000000460	NA	148.578792738572	86.6363793115826	101.231014192516
9	ENSG00000000938	NA	2134.7013294066	1614.11825741222	6782.47795089856
10	ENSG00000000971	NA	6.26537077813257	2.06277093599006	2.6293769920134
11	ENSG00000001036	NA	107.406356196558	111.389630543463	157.76261952090[illegible]
12	ENSG00000001084	NA	247.929672220389	390.895092370117	210.350159361072
13	ENSG00000001167	NA	351.755816543728	257.846366998758	269.511141681373
14	ENSG00000001460	NA	48.3328602884512	45.3809605917813	74.9372442723818
15	ENSG00000001461	NA	508.390085997042	450.715449513828	590.295134707008
16	ENSG00000001497	NA	412.619418388445	414.616958134002	395.721237298016
17	ENSG00000001561	NA	239.874195505647	369.235997542221	437.791269170231
18	ENSG00000001617	NA	0	3.09415640398509	3.9440654880201
19	ENSG00000001626	NA	0	1.03138546799503	9.20281947204689
20	ENSG00000001629	NA	848.51021395281	877.709033263771	1020.198272901 2
21	ENSG00000001630	NA	332.959704209331	354.796600990291	499.581628482546



Expression matrix in .gct format

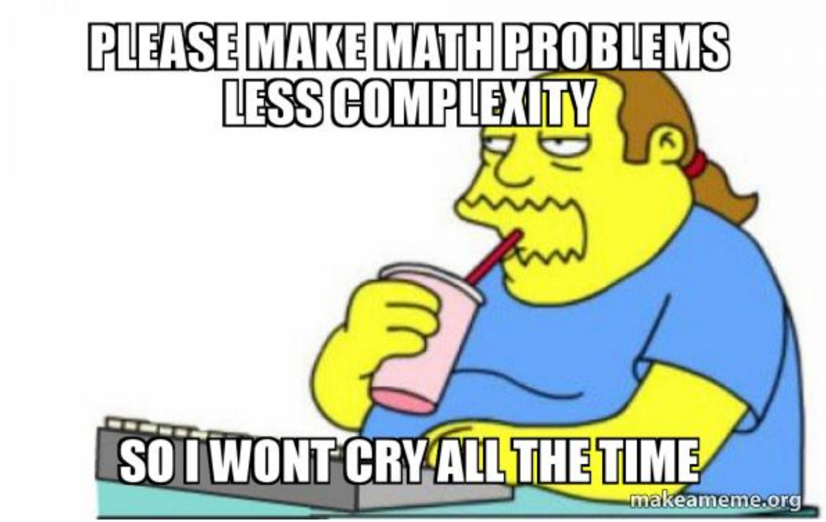
Metadata_gsea

Analysis customization

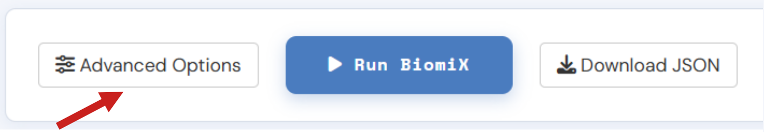
Just running default parameters can not be sufficient,

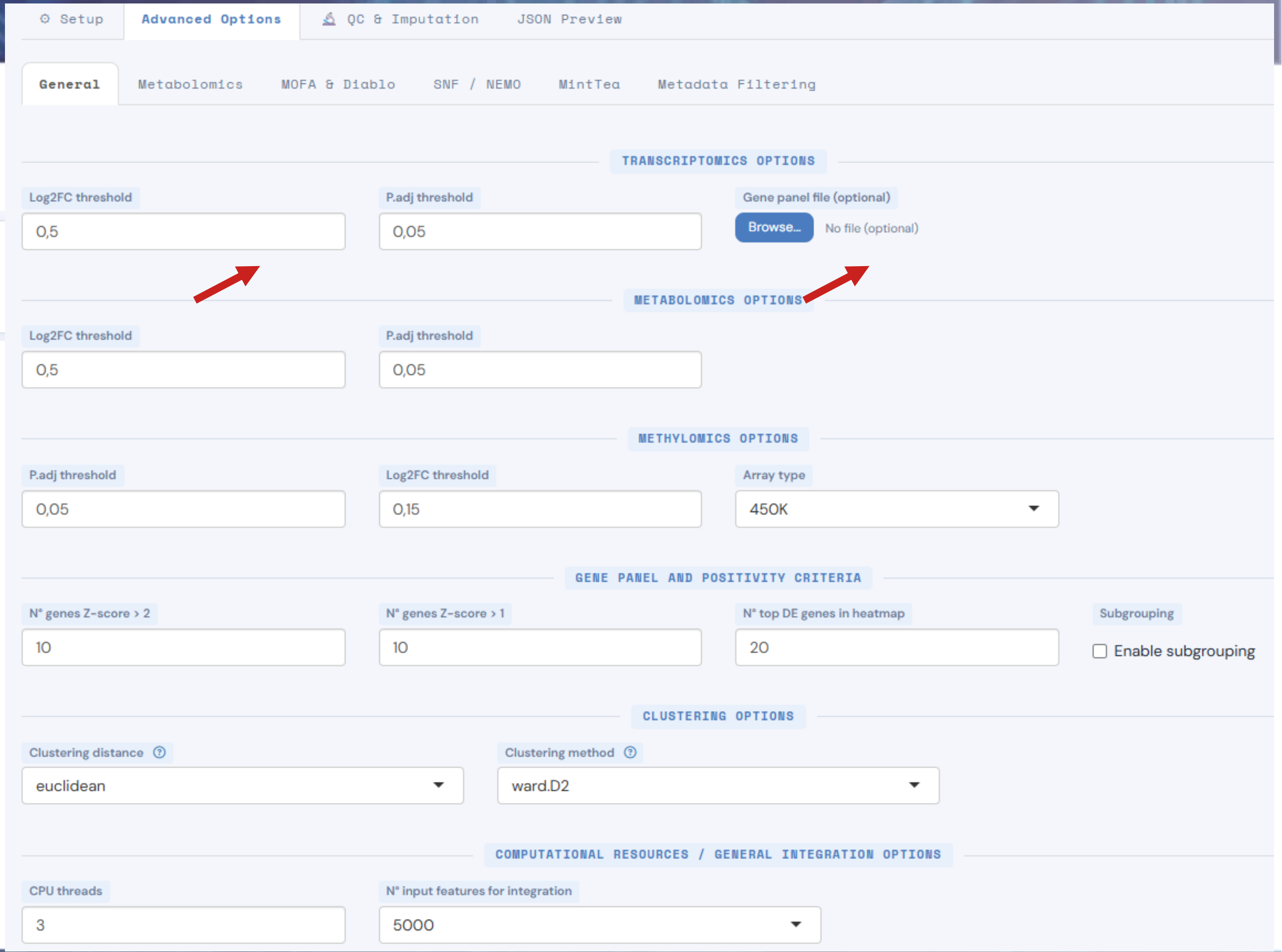
What if I want

- Increase the log2FC threshold
- Select only Male patients
- Stratify my patients by IFN signature

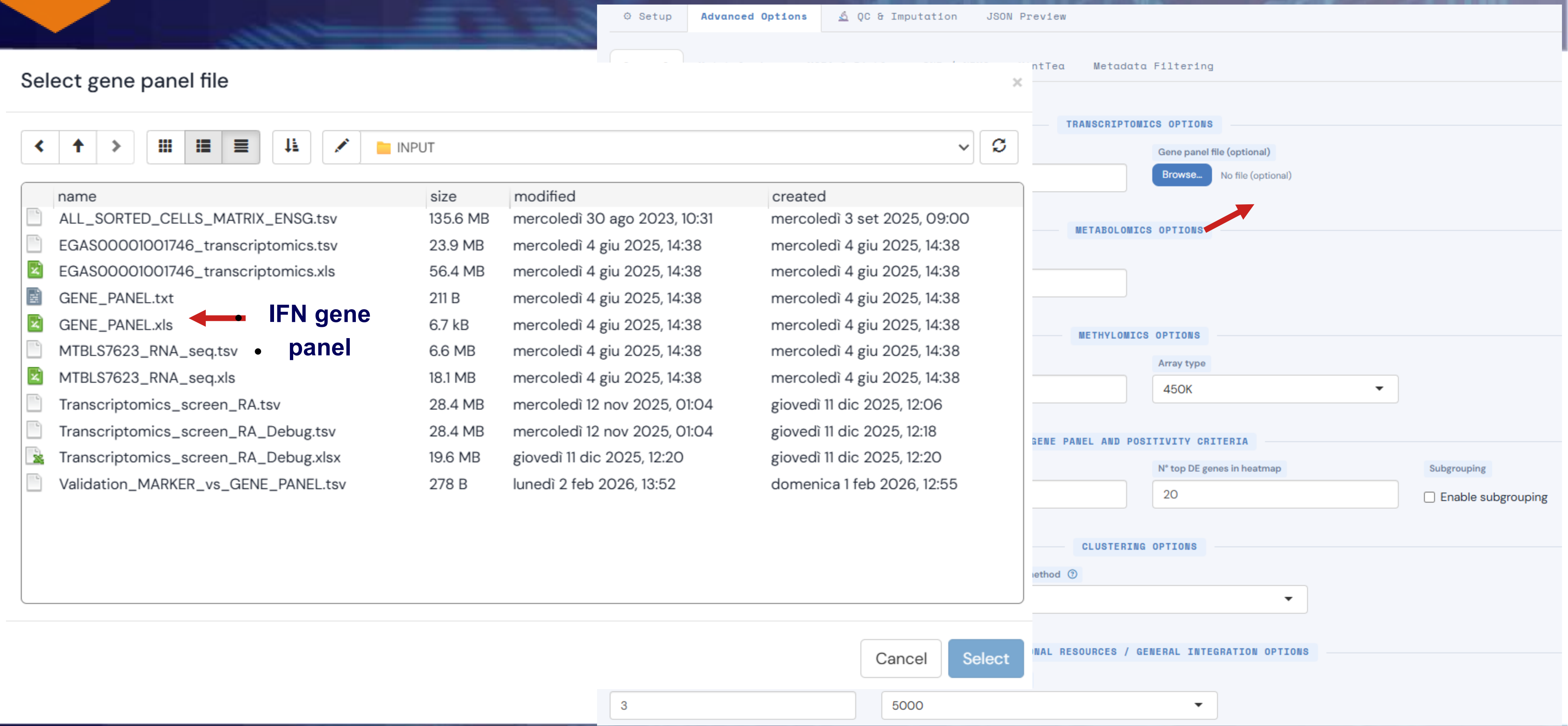


Analysis customization

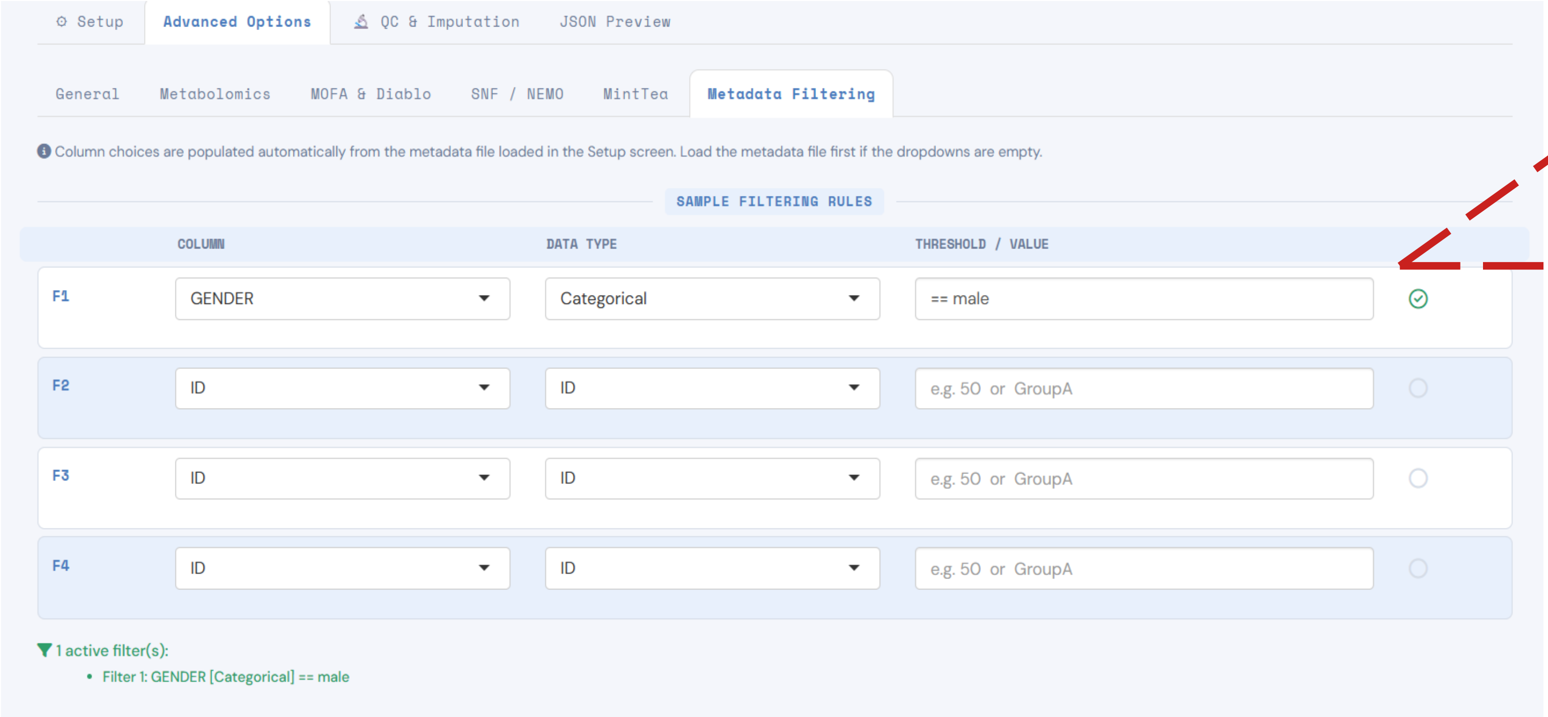




- Log2 fold change
- Pvalue
- N* gene in the heatmap
- Sample selection by gene panel


Setup
Advanced Options
QC & Imputation
JSON Preview
Metadata Filtering
TRANSCRIPTOMICS OPTIONS
Gene panel file (optional)
Browse...
No file (optional)
METABOLOMICS OPTIONS
METHYLOMICS OPTIONS
Array type
450K
GENE PANEL AND POSITIVITY CRITERIA
N° top DE genes in heatmap
20
Subgrouping
Enable subgrouping
CLUSTERING OPTIONS
NAL RESOURCES / GENERAL INTEGRATION OPTIONS
3
5000
Select gene panel file
INPUT
name
size
modified
created
ALL_SORTED_CELLS_MATRIX_ENSG.tsv 135.6 MB mercoledì 30 ago 2023, 10:31 mercoledì 3 set 2025, 09:00
EGAS00001001746_transcriptomics.tsv 23.9 MB mercoledì 4 giu 2025, 14:38 mercoledì 4 giu 2025, 14:38
EGAS00001001746_transcriptomics.xls 56.4 MB mercoledì 4 giu 2025, 14:38 mercoledì 4 giu 2025, 14:38
GENE_PANEL.txt 211 B mercoledì 4 giu 2025, 14:38 mercoledì 4 giu 2025, 14:38
GENE_PANEL.xls 6.7 kB mercoledì 4 giu 2025, 14:38 mercoledì 4 giu 2025, 14:38
MTBLS7623_RNA_seq.tsv 6.6 MB mercoledì 4 giu 2025, 14:38 mercoledì 4 giu 2025, 14:38
MTBLS7623_RNA_seq.xls 18.1 MB mercoledì 4 giu 2025, 14:38 mercoledì 4 giu 2025, 14:38
Transcriptomics_screen_RA.tsv 28.4 MB mercoledì 12 nov 2025, 01:04 giovedì 11 dic 2025, 12:06
Transcriptomics_screen_RA_Debug.tsv 28.4 MB mercoledì 12 nov 2025, 01:04 giovedì 11 dic 2025, 12:18
Transcriptomics_screen_RA_Debug.xlsx 19.6 MB giovedì 11 dic 2025, 12:20 giovedì 11 dic 2025, 12:20
Validation_MARKER_vs_GENE_PANEL.tsv 278 B lunedì 2 feb 2026, 13:52 domenica 1 feb 2026, 12:55
IFN gene panel
Cancel
Select

Analysis customization



Categorical

== egual to

!= different to

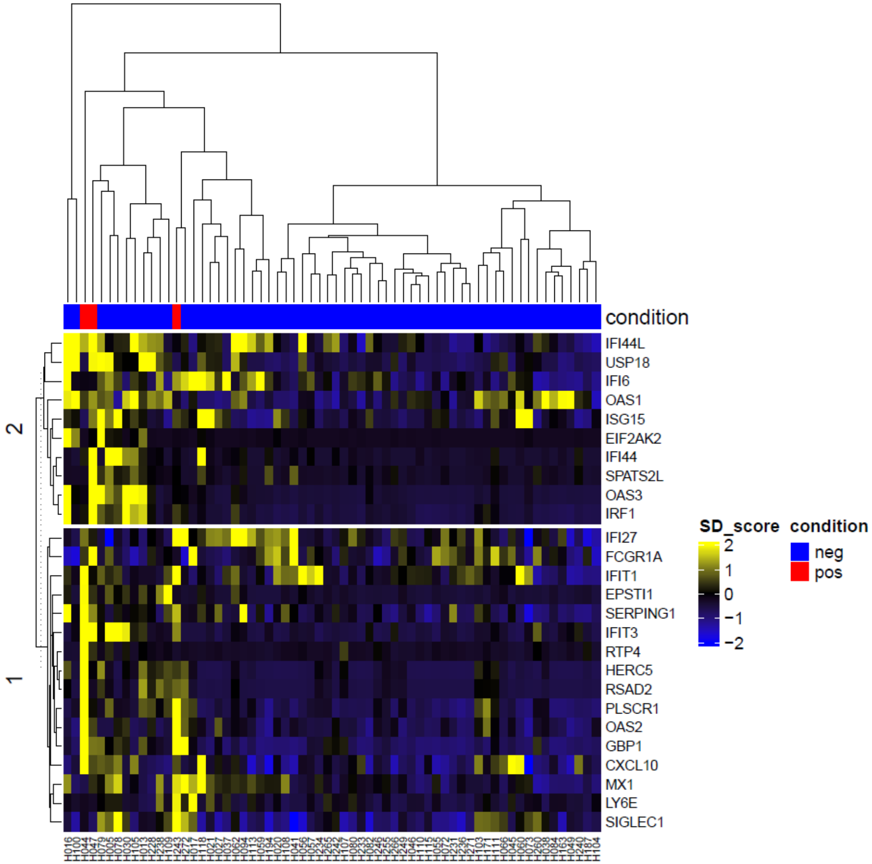
Numerical

> (>=) major of

< (<=) minor of

== egual to

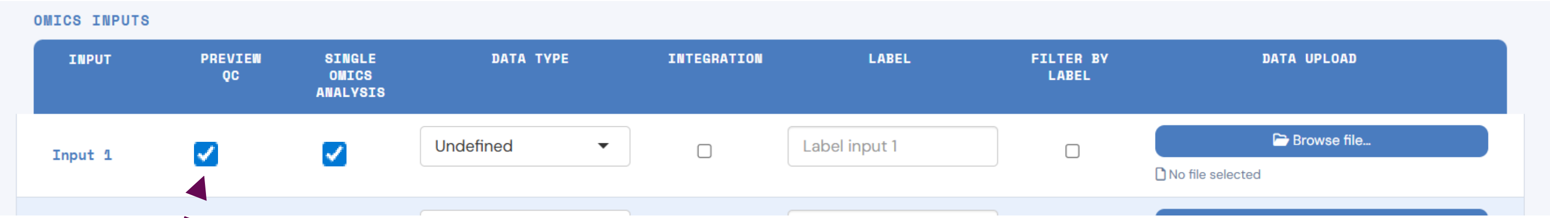
Always space*



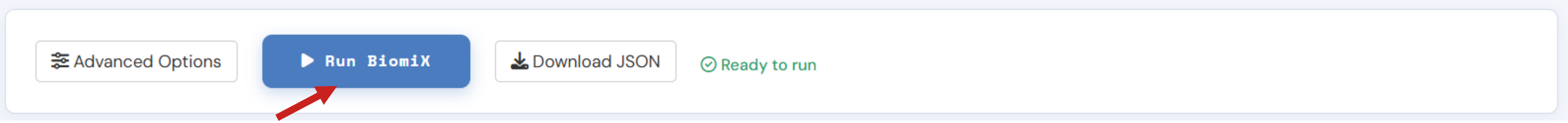
Directory:
\Integration\INPUT\RNA_mutated_vs_unmutated

Conclusion: IFN cannot discriminate CLL

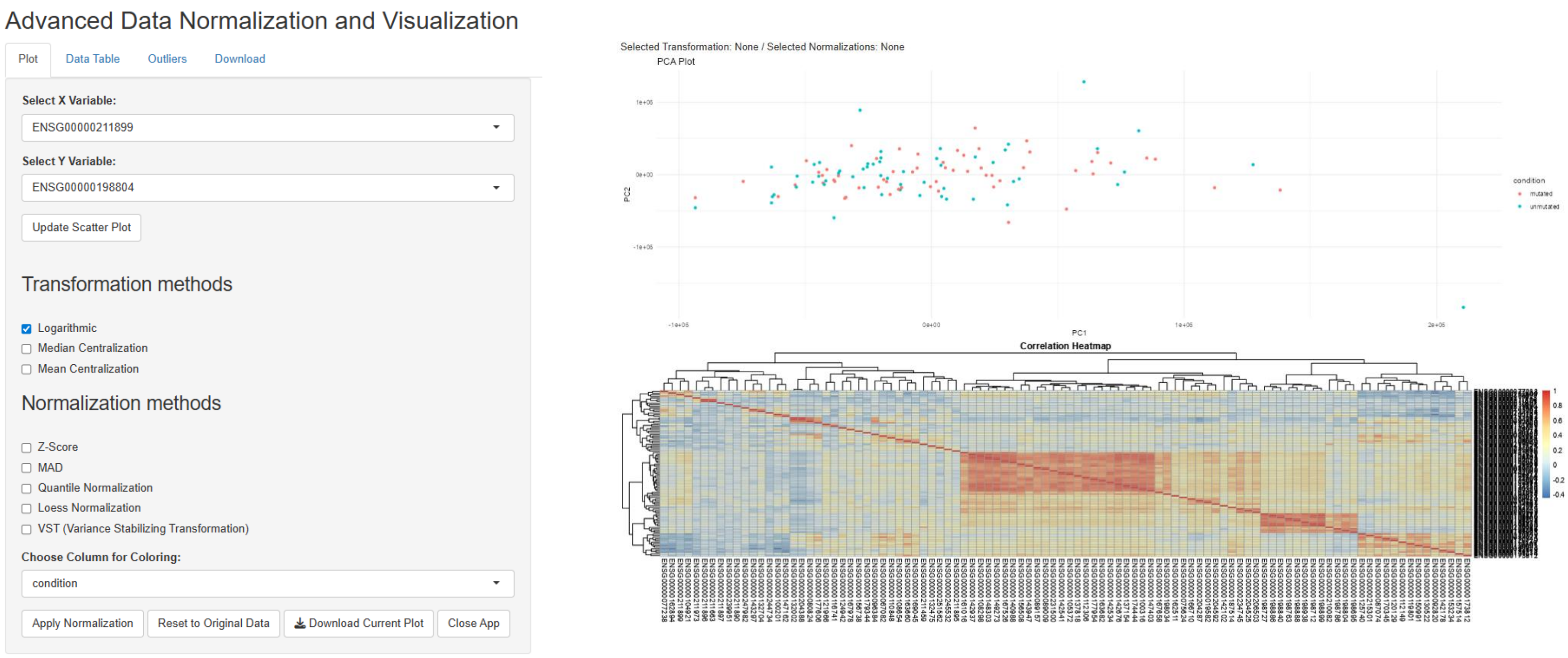
Other functions?



Normalization and interactive visualisation of the data before the analysis (Shiny app).



Same structure, different omics



Same structure, different omics

Advanced Data Normalization and Visualization

Plot | Data Table | Outliers | Download

Select X Variable:

ENSG00000211899

Select Y Variable:

ENSG00000198804

Update Scatter Plot

Transformation methods

- [x] Logarithmic
- [ ] Median Centralization
- [ ] Mean Centralization

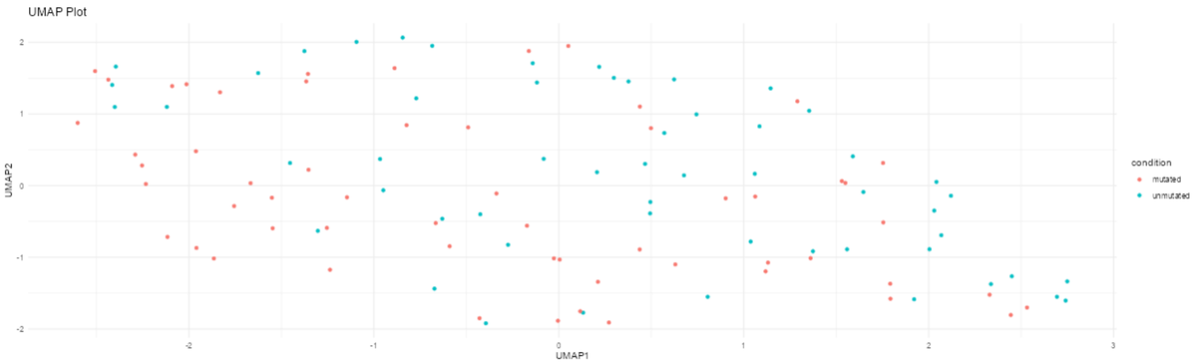
Normalization methods

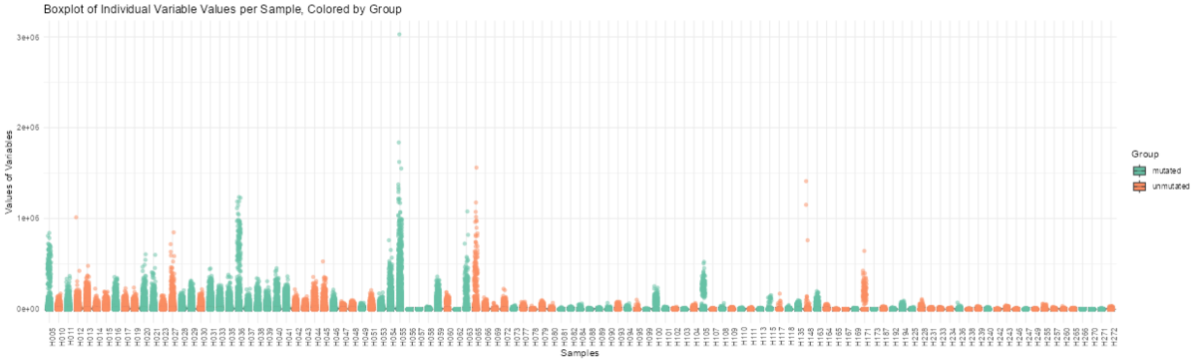
- [ ] Z-Score
- [ ] MAD
- [ ] Quantile Normalization
- [ ] Loess Normalization
- [ ] VST (Variance Stabilizing Transformation)

Choose Column for Coloring:

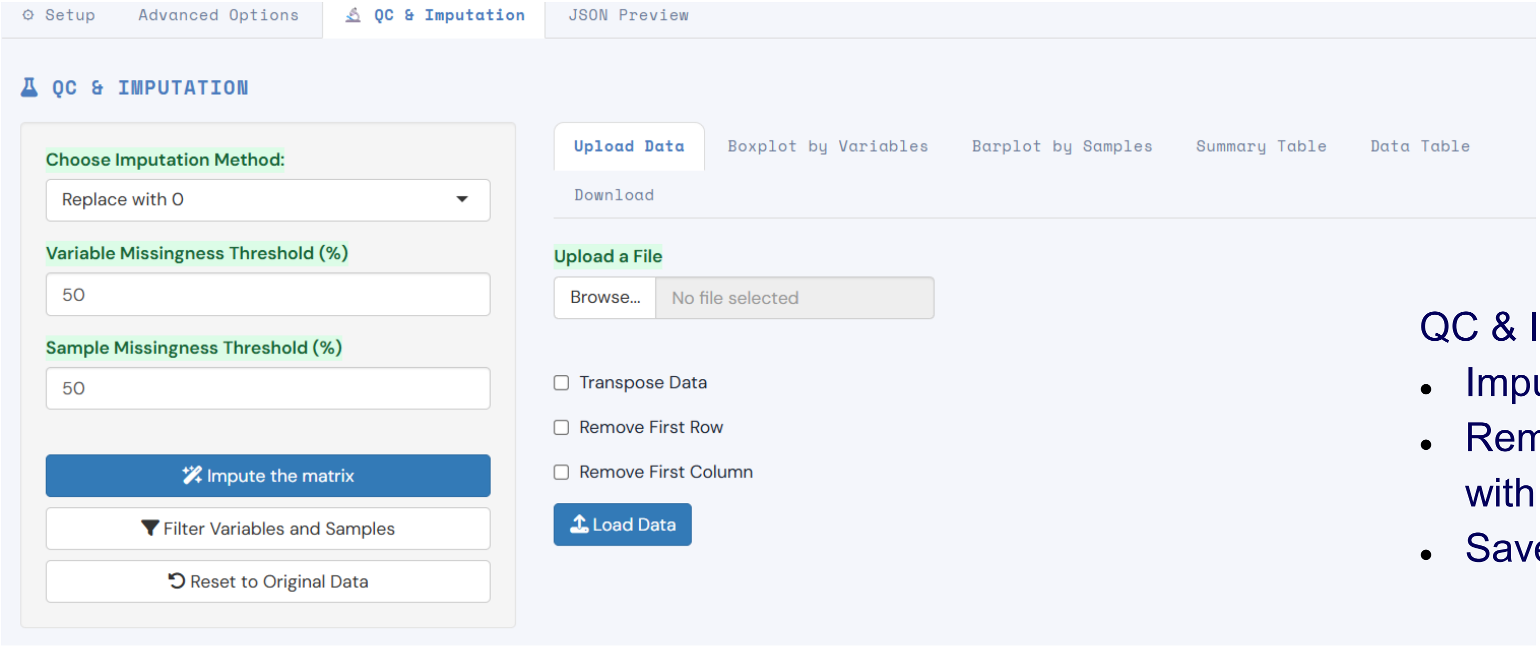
condition

Apply Normalization | Reset to Original Data | Download Current Plot | Close App





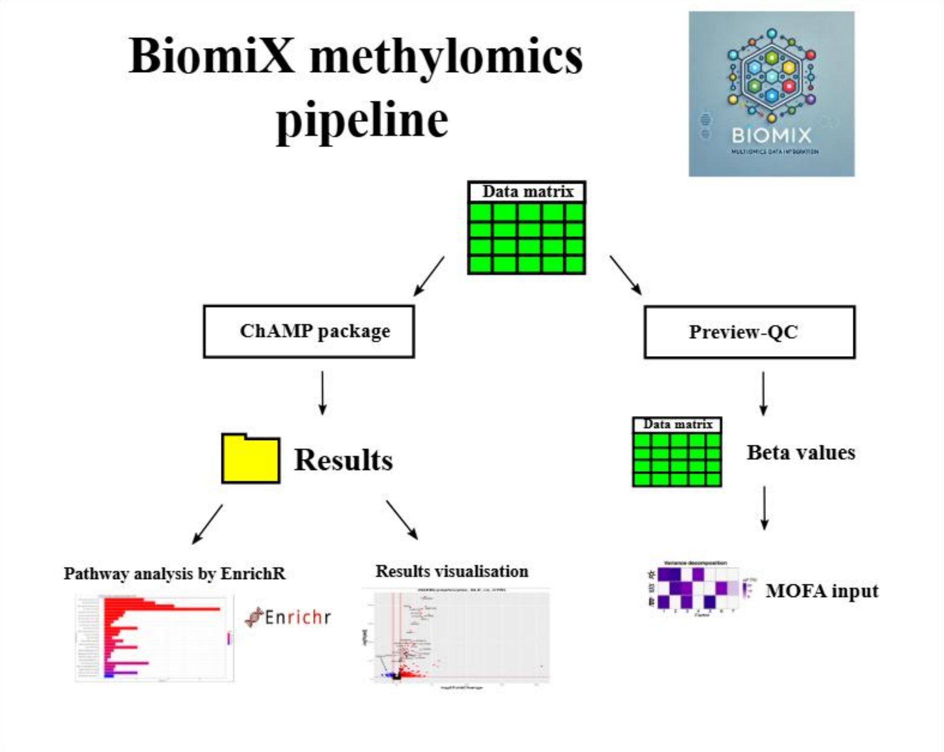
Other functions?



QC & Imputation

- Imputation of missing data
- Removal of sample and variables with NaN data
- Save of the new table

Same structure, different omics



1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder. (Metadata slot)

2) Load EGAS00001001746_methylomics.tsv from Methylomics folder. (Input 2)

* To avoid re-running also the transcriptomics analysis deselect "single omics" from the slot 1.

OMICS INPUTS

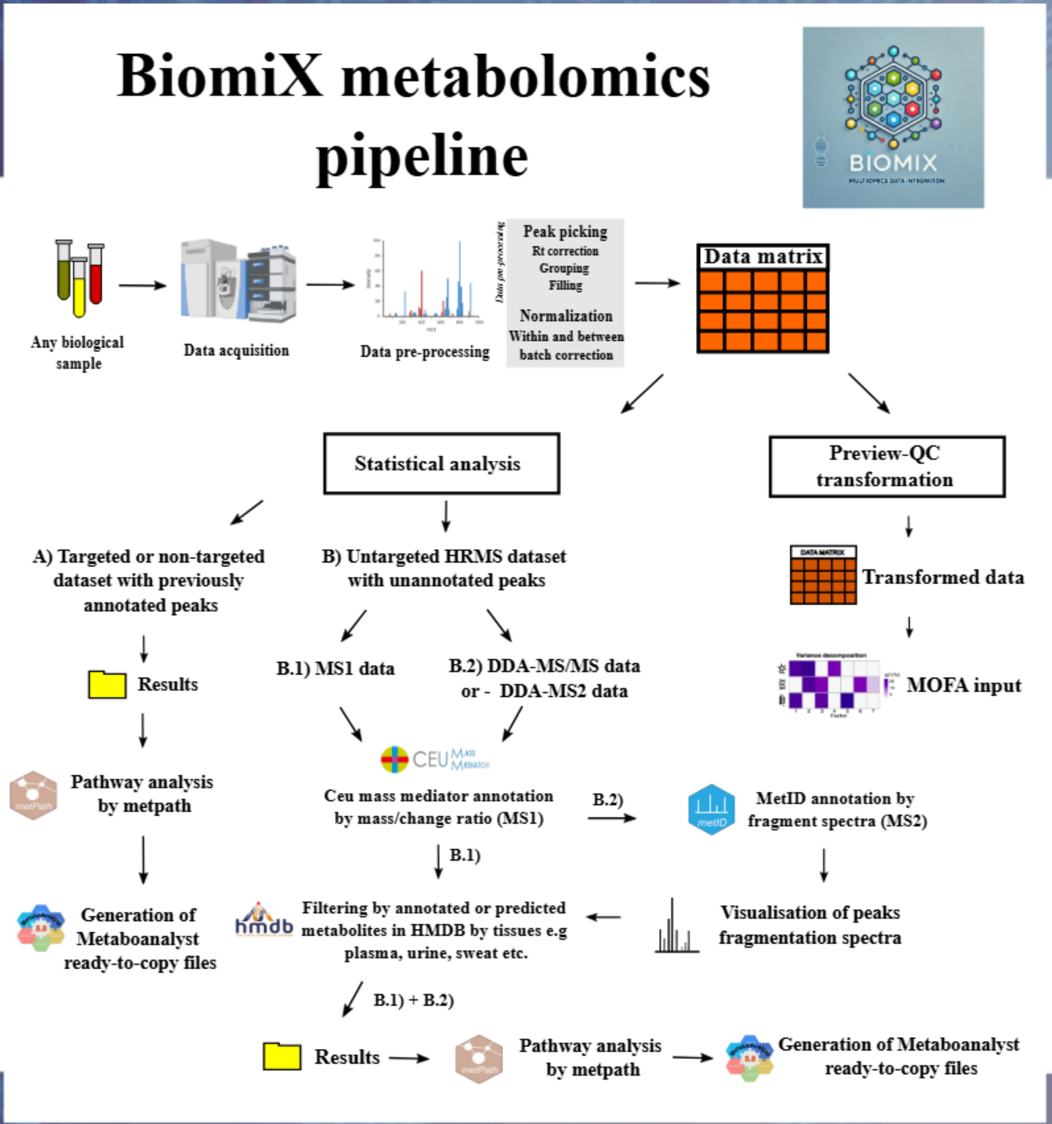
INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL
Input 1	☐	☐	Transcriptomics	☐	RNA
Input 2	☐	☑	Methylomics	☐	Methylomics

Metabolomics analysis PTB

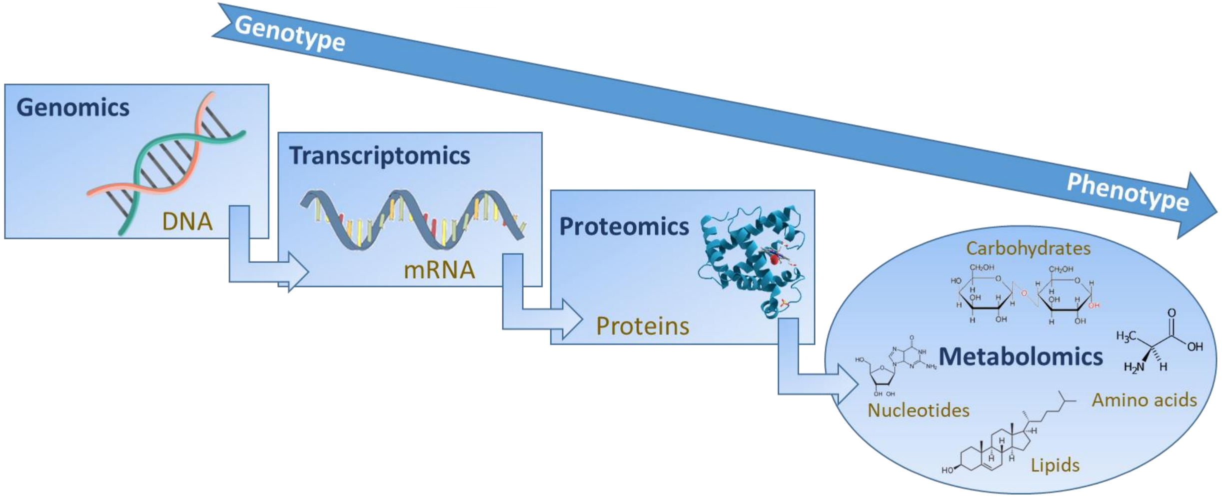
Álvaro Fernández-Ochoa

University of Granada, Spain

UNIVERSIDAD DE GRANADA



What is metabolomics?



Molecular weight < 1500 Da

metabolite levels closely associate with phenotype

metabolomics

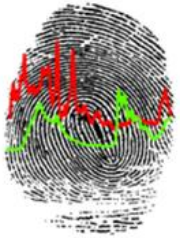
analysis of low molecular weight metabolites in biological fluids

"Comprehensive profiling of metabolites *key* to obtain in depth insight into disease biochemistry"

Metabolomics approaches

Targeted metabolomics


Untargeted metabolomics

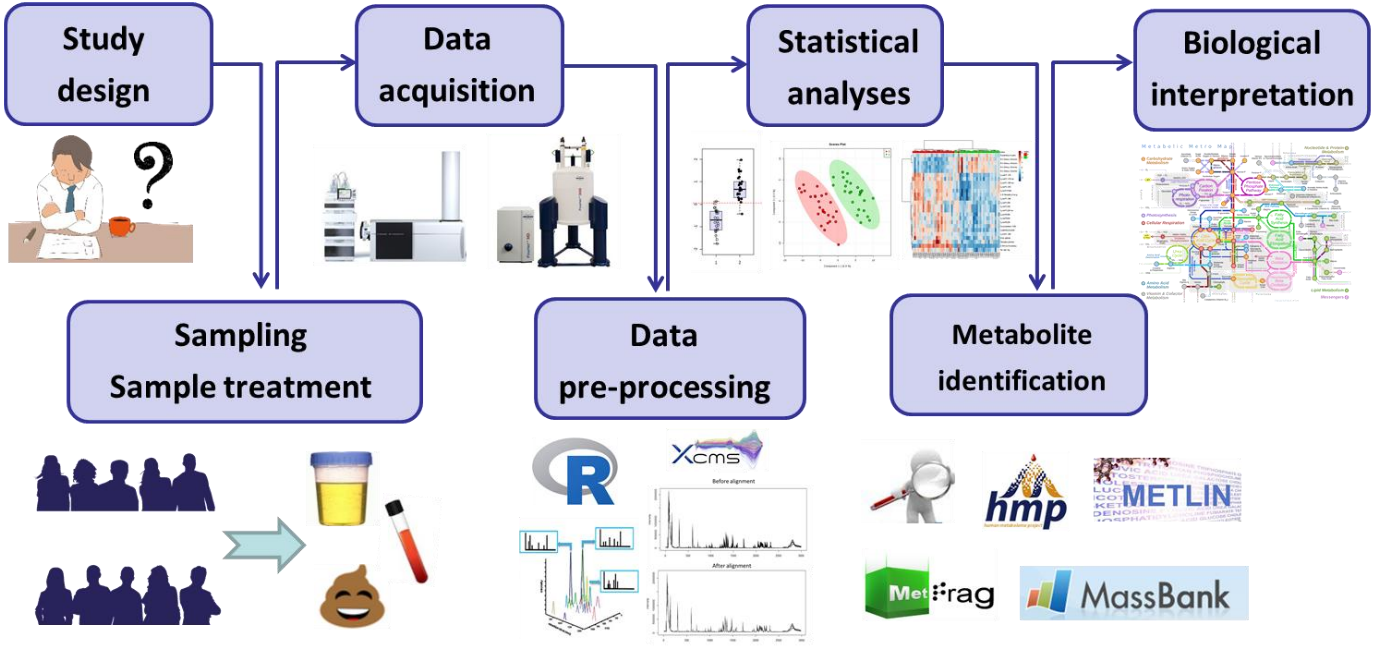



BIOMIX supports data from both approaches

BIOMIX

Predefi	erage
Previously known metabolites	Identification of signals of interest
Absolute quantification	Relative quantification
Higher sensitivity	Lower sensitivity
Simplicity in data processing	Complexity in data processing

Metabolomics workflows



COMPUTATIONAL TOOLS

From raw to processed data

Data acquisition

- Data conversion from vendor-specific to open formats (mzML, abf, etc)

Pre-processing

- Mass/peak detection
- Noise filtering
- Peak deconvolution
- Isotope filtering
- Alignment
- Gap-filling
- Duplicate filtering

Normalization

- Drift Correction
- Batch effect correction
- Low-quality feature identification
- Missing value imputation
- Transformation
- Scaling

Statistics

Annotation

Biological interpretation

COMPUTATIONAL TOOLS

From raw to processed data

✕ Not covered by BiomiX

Data acquisition

- Data conversion from vendor-specific to open formats (mzML, abf, etc)

Pre-processing

- Mass/peak detection
- Noise filtering
- Peak deconvolution
- Isotope filtering
- Alignment
- Gap-filling
- Duplicate filtering

Normalization

- Drift Correction
- Batch effect correction
- Low-quality feature identification

Sophisticated QC-based methods (drift/batch-effect correction, feature filtering) are NOT covered by BiomiX in the current version

- Missing value imputation
- Transformation
- Scaling

Transformation, scaling, NA imputation & several normalization methods: covered

Statistics

Annotation

Biological interpretation

✓ Covered by BiomiX

BiomiX requires an already pre-processed data matrix as input.
Raw data pre-processing and QC-based drift/batch-effect correction must be performed beforehand, following proper QA/QC criteria (especially in untargeted metabolomics).

Data pre-processing

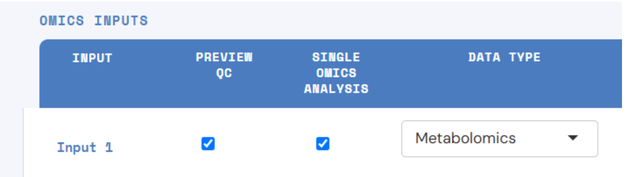
Data processing concepts	Definition
Preprocessing	
Data conversion	Conversion of vendor-specific raw data formats into open formats, such as mzML, mzXML, or ABF, enabling interoperability across analysis software
Peak picking	Detection of ion signals above a defined noise threshold, enabling the detection of potential metabolite signals
Noise filtering	Removal of low-intensity signals that do not represent true molecular features
Peak deconvolution	Separation of overlapping chromatographic peaks corresponding to different compounds that co-elute during chromatographic separation
Alignment	Correction of retention-time shifts across samples to ensure that the same molecular features are matched across datasets
Peak integration	Calculation of peak areas or intensities to estimate metabolite abundance across samples
Grouping of related features	Clustering of isotopes, adducts, in-source fragments, and multimers to represent a single metabolite as one molecular feature

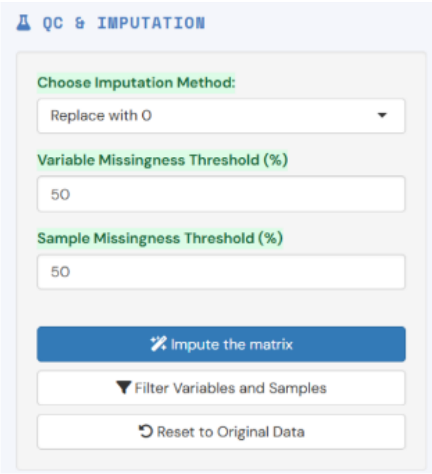
Data pre-processing

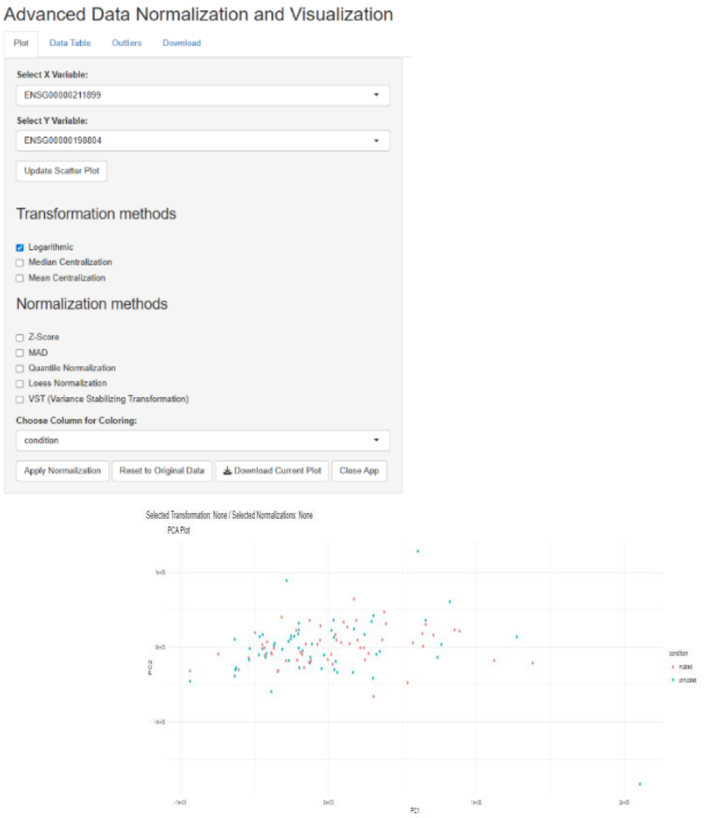
Data processing concepts	Definition
Data correction	
Quality control assessment	Evaluation of analytical reproducibility and instrument stability using QC samples
Blank subtraction/ contaminant removal	Removal of background signals, contaminants, or solvent-derived peaks
Drift correction	Correction of systematic signal variation caused by instrumental fluctuations over time during a single analytical batch or run sequence, often monitored using QC samples.
Batch normalization	Adjustment of signal intensities to reduce inter-batch variability when samples are analyzed across multiple experimental runs or instruments, ensuring comparability across batches.
Missing value imputation	Estimation or replacement of missing values to allow statistical analysis.
Data transformation	Mathematical transformation of signal intensities (e.g., log, square root, cube root) to stabilize variance, reduce skewness, and make data more normally distributed for statistical analysis.
Scaling	Adjustment of metabolite intensities to standardize variance across features (e.g., autoscaling, Pareto scaling), to give equal importance to low- and high-intensity metabolites in multivariate statistical analysis.

Data pre-processing

Normalization and visualisation of the data before the analysis (Shiny app).





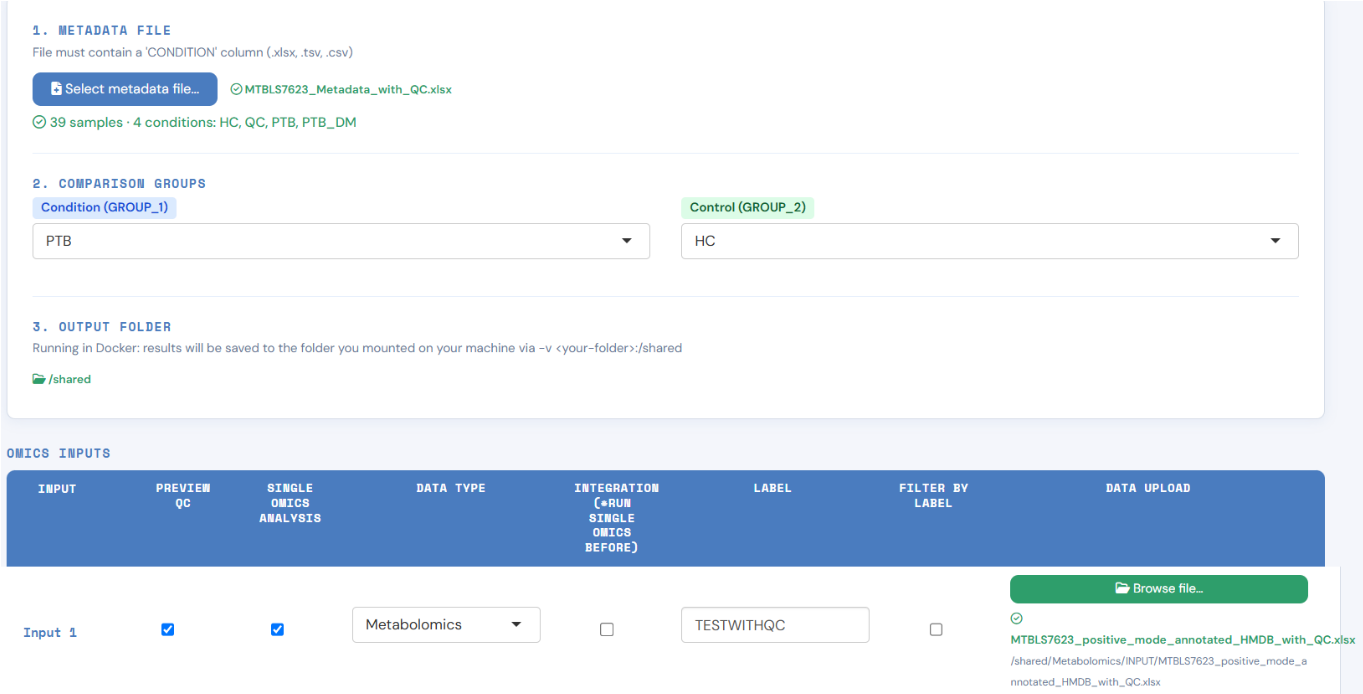


BiomiX supports datasets with QC samples, enabling data quality assessment based on their clustering in PCA.

Data pre-processing

Normalization and visualisation of the data before the analysis (Shiny app).

BiomiX supports datasets with QC samples, enabling data quality assessment based on their clustering in PCA.



Data pre-processing

Transformation methods

- [x] Logarithmic
- [ ] Median Centralization
- [ ] Mean Centralization

Normalization methods

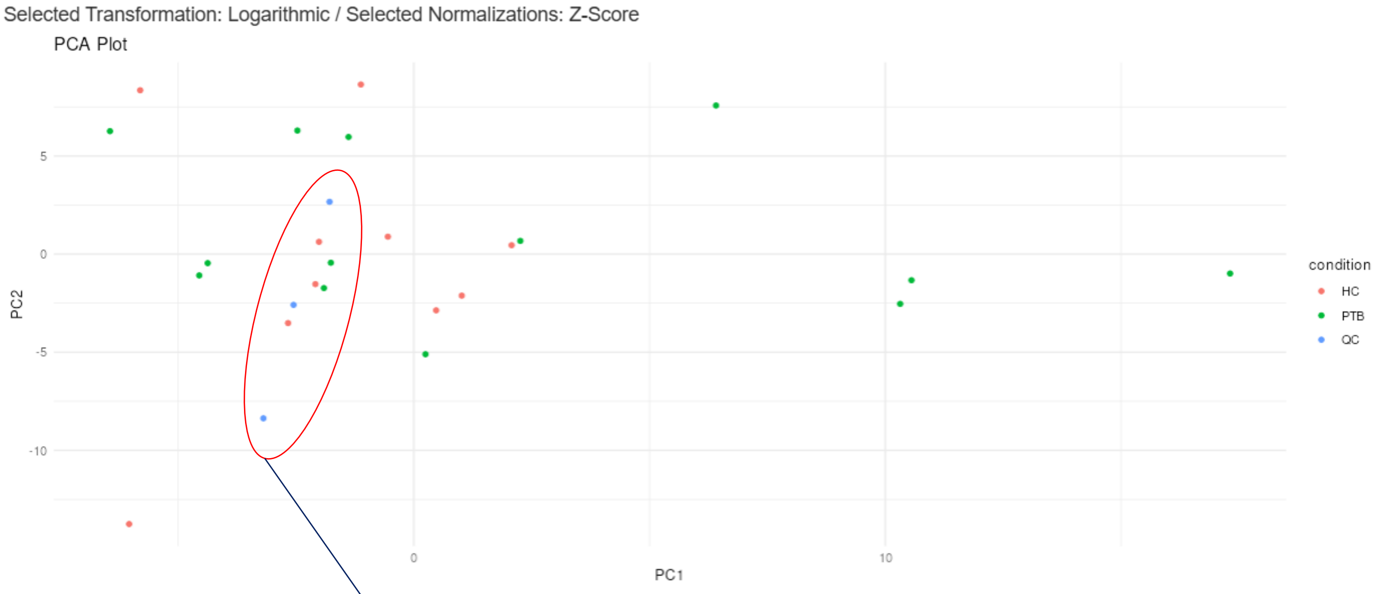
- [x] Z-Score
- [ ] MAD
- [ ] Quantile Normalization
- [ ] Loess Normalization
- [ ] VST (Variance Stabilizing Transformation)

Choose Column for Coloring:

condition

Apply Normalization | Reset to Original Data

Download Current Plot | Close App

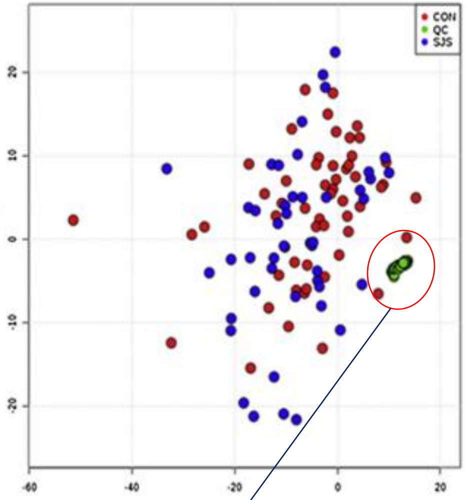


Not Good QC grouping..

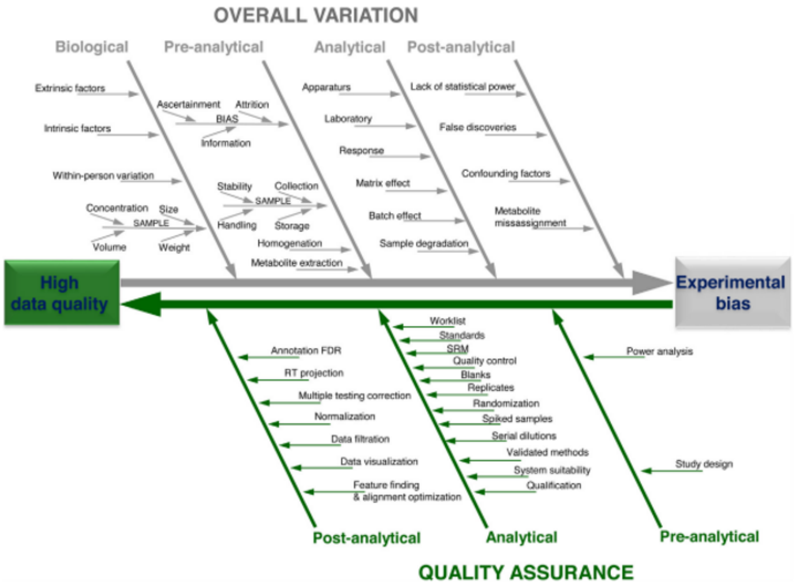
QA/QC - Metabolomics

Quality Control and Quality Assurance (QC/QA)

- **QA** defines processes planned and performed to fulfill defined quality requirements
- **QC** comprises measures to report whether these quality requirements have been met



Good QC grouping..



Journal of Pharmaceutical and Biomedical Analysis 147 (2018) 149–173

Contents lists available at ScienceDirect

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journal homepage: www.elsevier.com/locate/jpba

Review

Quality assurance procedures for mass spectrometry untargeted metabolomics. a review

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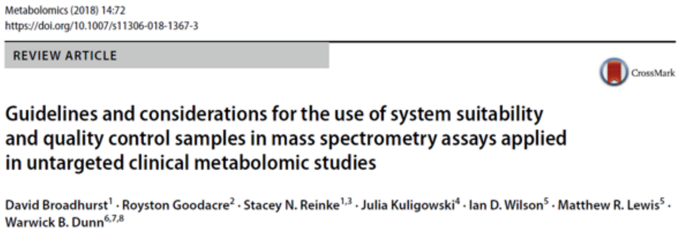


Metabolomics (2018) 14:72

https://doi.org/10.1007/s11306-018-1367-3

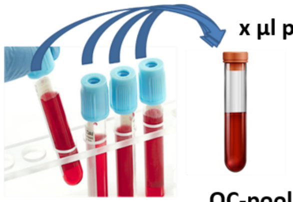
REVIEW ARTICLE

Guidelines and considerations for the use of system suitability and quality control samples in mass spectrometry assays applied in untargeted clinical metabolomic studies

David Broadhurst[1] · Royston Goodacre[2] · Stacey N. Reinke[1,3] · Julia Kuligowski[4] · Ian D. Wilson[5] · Matthew R. Lewis[5] · Warwick B. Dunn[6,7,8]

QA/QC - Metabolomics

Types of QC samples



x µl per sample

QC-pooled sample

- **Intra-study QC (Pooled QC)** →
    - Using all samples
    - Using a subset of samples

    - Pooled before extraction (fluids,...)
    - Pooled after extraction (tissue samples,...)

- **Intra-lab QC**
- **System Suitability QC**
- **Inter-lab QC**
    - *QC-pooled across multiple studies following a defined SOP*
    - *Standard Reference Material (SRM) sample*
    - *Reference Material sample*
    - *Synthetic mixture of authentic chemical standards*
- **Process blank**

Uses of QC Samples

Perspective | December 06, 2023 Open Access

Current Practices in LC-MS Untargeted Metabolomics: A Scoping Review on the Use of Pooled Quality Control Samples

Corey D. Broeckling *; Richard D. Beger; Leo L. Cheng; Raquel Cumeras; Daniel J. Cuthbertson; Surendra Dasari; W. Clay Davis; Warwick B. Dunn; Anne Marie Evans; Alvaro Fernández-Ochoa; Helen Gika; Royston Goodacre; Kelli D. Goodman; Goncalo J. Gouveia; Ping-Ching Hsu; Jennifer A. Kirwan; Dritan Kodra; Julia Kuligowski; Renny Shang-Lun Lan; María Eugenia Monge; Laura W. Moussa; Sindhu G. Nair; Nichole Reisdorph; Stacy D. Sherrod; Candice Ulmer Holland; Dajana Vuckovic; Li-Rong Yu; Bo Zhang; Georgios Theodoridis *; Jonathan D. Mosley *

+ Author & Article Information

Anal. Chem. (2023) 95 (51): 18645–18654.
https://doi.org/10.1021/acs.analchem.3c02924 **Article history**

- Method optimization
- Validation
- SST
- System conditioning/Stabilization
- Assess data quality: check precision, linearity, filtering based on NA,
- Normalization (batch effects, ...)
- Be a representative sample to perform MS/MS análisis
- Detect contaminants (process blanks)

COMPUTATIONAL TOOLS

- **Data preprocessing**
- **Statistical analysis**
- Metabolite annotation
- Biological interpretation

Open-source platforms

OpenMS
MS-DIAL
Workflow4metabolomics
XCMS
mzmine

Vendor software

Met
Progenesis QI
Cd

Software and specialized platforms

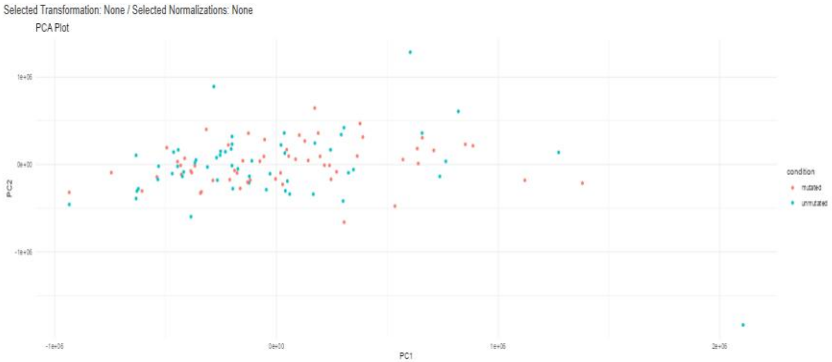
MetaboAnalyst 6.0
MATLAB
BIOMIX
MULTIOMICS DATA INTEGRATION

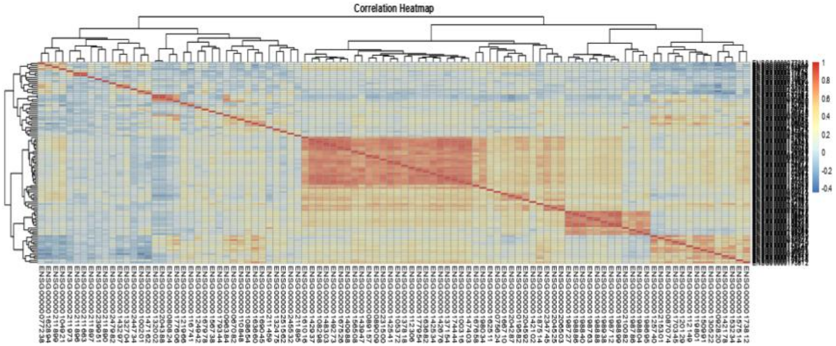
Open-source programming languages

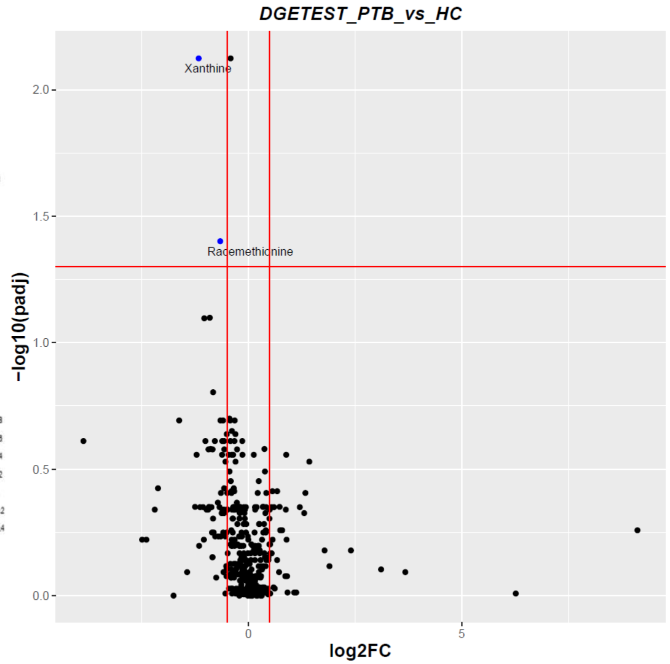
R
RStudio
jupyter
colab

Biomix – Statistics

- Log2 fold change
- Pvalue
- Volcano Plot
- PCA







Metabolite annotation

Compound annotation

Databases

LIPID MAPS, KEGG, FOOD3, PubChem, Phenol-Explorer, ChemSpider, CEU Mass Mediator, hmdb

BIOMIX
MULTIOMICS DATA INTEGRATION

Mass banks

MassBank
High Quality Mass Spectral Database

MoNA - MassBank of North America

Computational tools

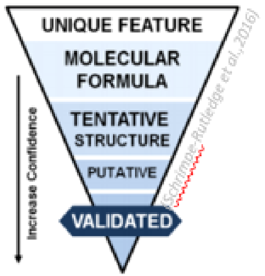
MS-FINDER, GNPS2, SIRIUS

+ ML and DL algorithms, such as convolutional neural networks (CNNs), to predict molecular fingerprints from MS/MS spectra

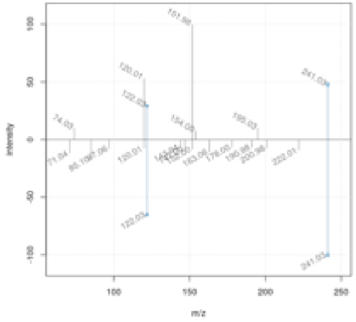
Critical step (main bottleneck) in untargeted metabolomics

- Comparison of *m/z*, isotopic pattern, RT, and MS/MS or MSn fragmentation spectra
- Report following the Metabolomics Standards Initiative (MSI) levels

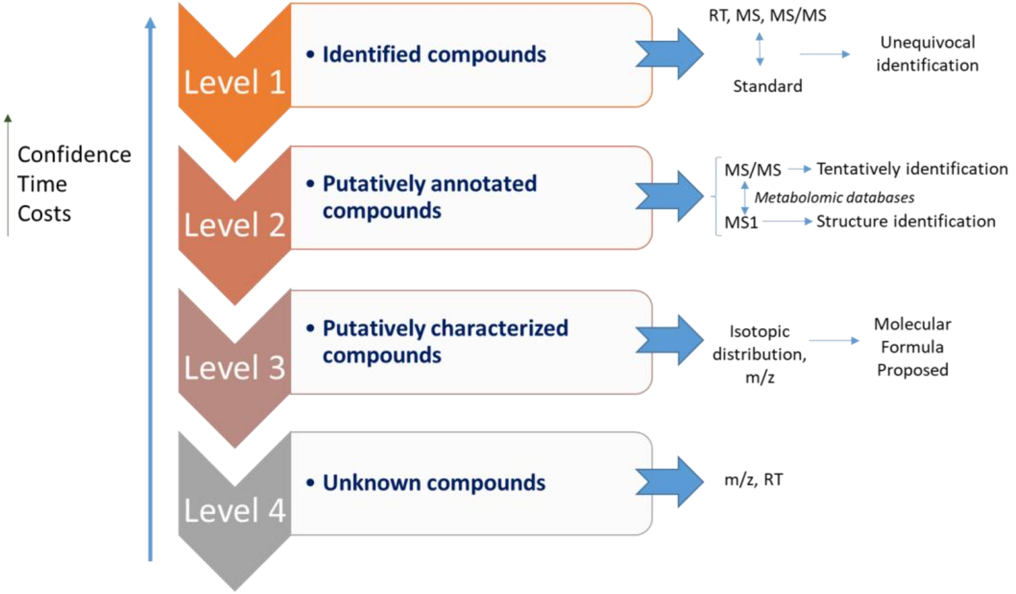
MSI levels



MS/MS spectra



Metabolomics analysis PTB



Same analysis different depth of annotation

Run 1) Already annotated dataset (e.g. Targeted)

Run 2) Level 2 annotation MS/MS + Level 3.

Run 3) Level 3 annotation (m/z) and retention time

Reisdorph, Nichole et al.. Clinical Metabolomics (2019)

Metabolomics analysis PTB

⚙ Setup | Advanced Options | QC & Imputation | JSON Preview

General | Metabolomics | MOFA & Diablo | SNF / NEMO | MintTea | Metadata Filtering

Annotation mode

- ○ Annotated (pre-annotated file with compound_name / HMDB / KEGG) → No annotation required (you have it already) — ***Targeted, or Untargeted with pre-annotated results***
- ◉ MS1 only (non-targeted m/z matching against databases) → Level 3 annotation by (m/z) and retention time
- ○ MS1 + MS2 (m/z matching + spectral matching) → Level 2 annotation by MS/MS + m/z and retention time

Untargeted metabolomics

1. METADATA FILE

File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file... MTBLS7623_Metadata.xls

39 samples · 3 conditions: HC, PTB, PTB_DM

1) Load MTBLS7623_Metadata.xlsx from Metadata folder. (Metadata slot)

2. COMPARISON GROUPS

Condition (GROUP_1): PTB

Control (GROUP_2): HC

3. OUTPUT FOLDER

Folder where BiomiX will save results and the JSON configuration file

Select folder... C:/Users/crist/Desktop/BiomiX2.5

2) Load MTBLS7623_positive_mode_annotated_HMDB.xls from Metabolomics folder. (Input 1)

	A	B	C	D
1	ID	HC_1	HC_10	HC_11
2	HMDB0012497	8958855	11731.87022	11096.62856
3	HMDB0000925	3062.01725	2260.162667	3635.334667
4	HMDB0034301	174354.5458	136631.2239	116859.7333
5	HMDB0034301	50937.20491	44507.57589	50626.45636
6	HMDB0031666	9584.321591	10557.13338	9649.629818
7	HMDB0000097	81410.99886	82769.48167	68992.30848

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☑	Metabolomics	☐	TEST	☐	Browse file... MTBLS7623_positive_mode_annotated_HMDB.xls C:/Users/crist/Desktop/BiomiX2.5/Metabolomics/INPUT/MTBLS7623_positive_mode_annotated_HMDB.xls
Input 2	☐	☐	Undefined	☐	Label input 2	☐	Browse file... No file selected

Metabolomics analysis PTB

1) In Advance Options select the “Annotated mode”

Metabolomics analysis PTB

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_DOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABDOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB

1) Open:
Metabolomics/OUTPUT/TEST_PTB_vs_HC

Metabolomics analysis PTB

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
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TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB

Only significant negative hit

Only significant positive hit

Results table

	A	B	C	D	E	F	G	H	I
1	shapiro_pv	p_val	log2FC	NAME	padj	Name	chemical_	average_m	kegg_id
2	0.082004	2.67E-05	-1.16687	HMDB000(	0.007518	Xanthine	C5H4N4O:	152.1109	C00385
3	0.778672	0.000297	-0.66172	HMDB003:	0.039712	Racemethi	C5H11NO:	149.211	C01733

Metabolomics analysis PTB

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
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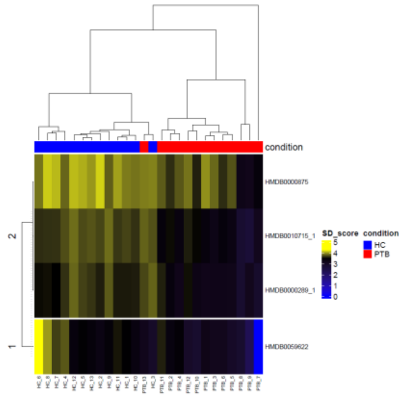
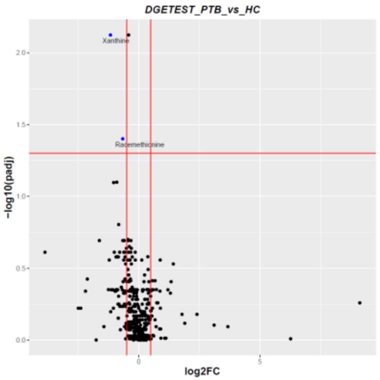
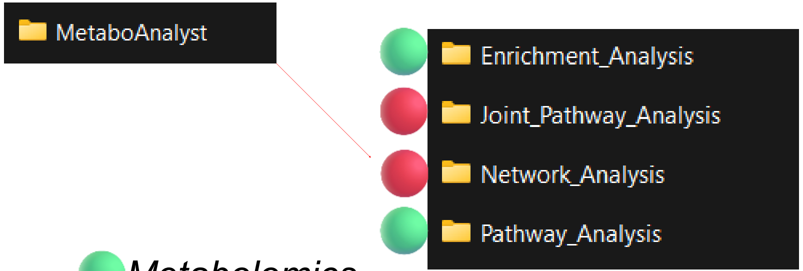
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB





From BiomiX to biological interpretation with MetaboAnalyst



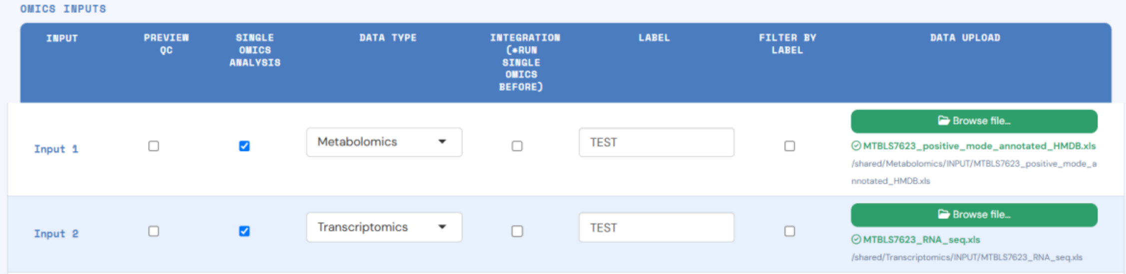
Metabolomics
Metabolomics + transcriptomics

https://www.metaboanalyst.ca/

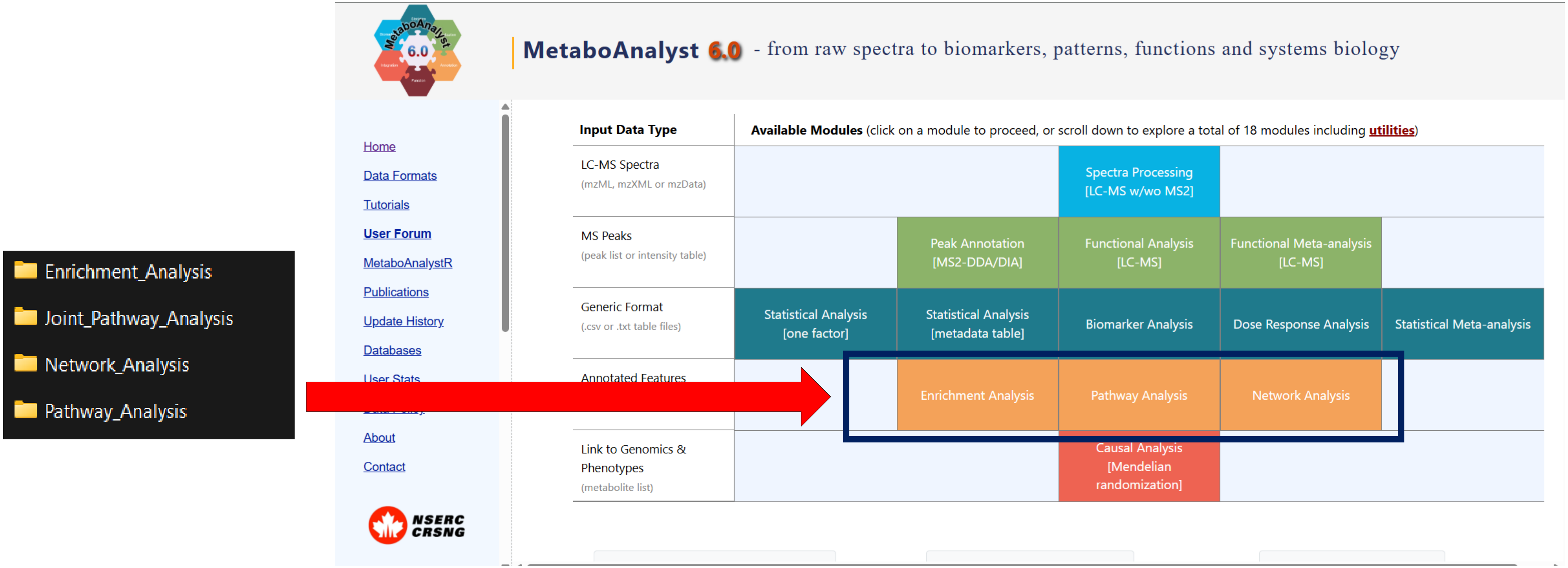
BiomiX automatically generates analysis-ready **outputs that can be directly used in MetaboAnalyst for downstream functional and pathway interpretation**.

Key advantage: BiomiX automatically prepares the required input files, reducing manual data handling and reformatting.

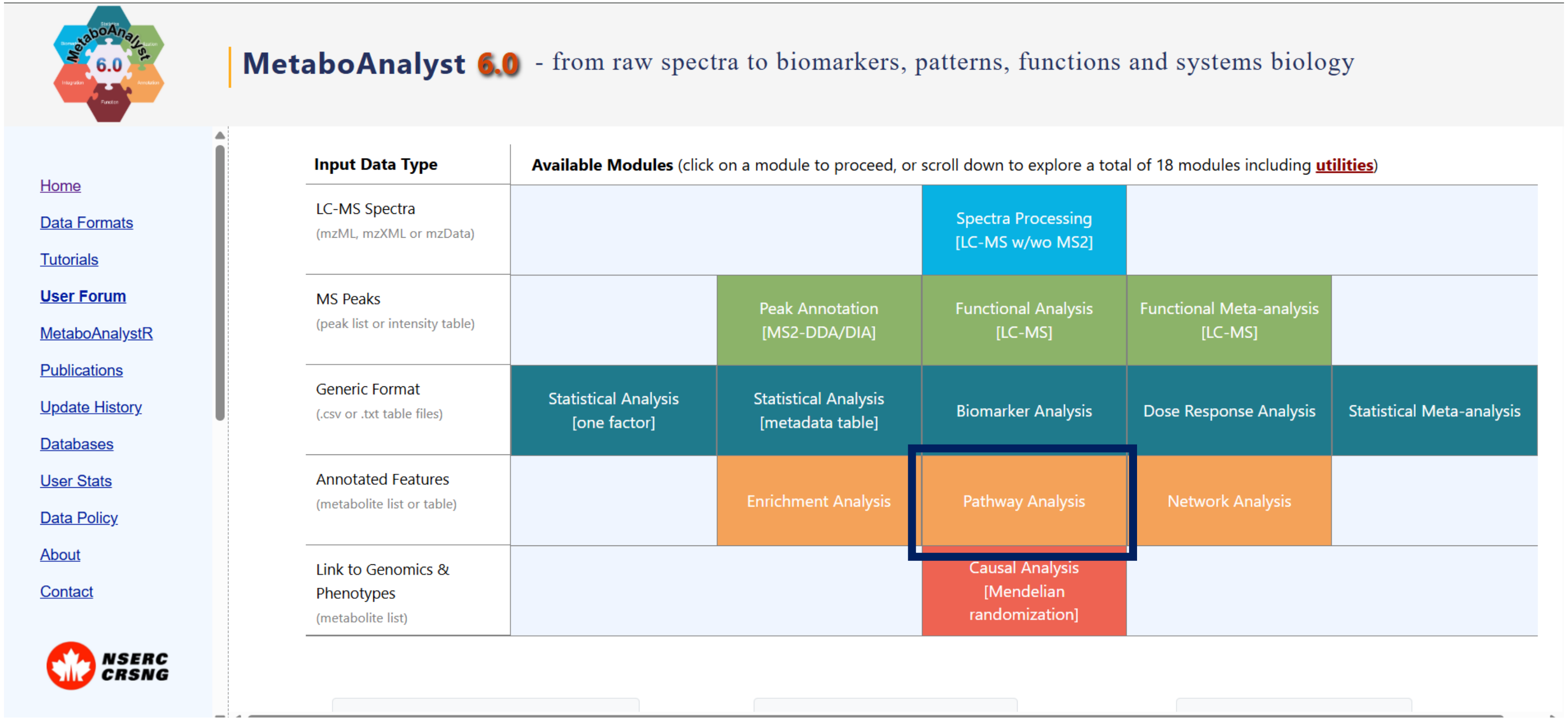
Multi-omics ready: when transcriptomics data are also processed in BiomiX, combined outputs are generated for Joint Pathway and Network analyses.

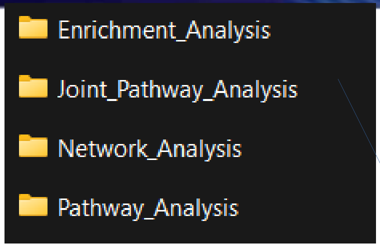


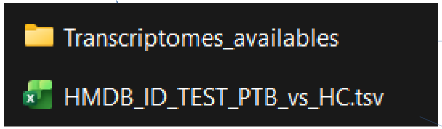
Example usage on Metaboanalyst

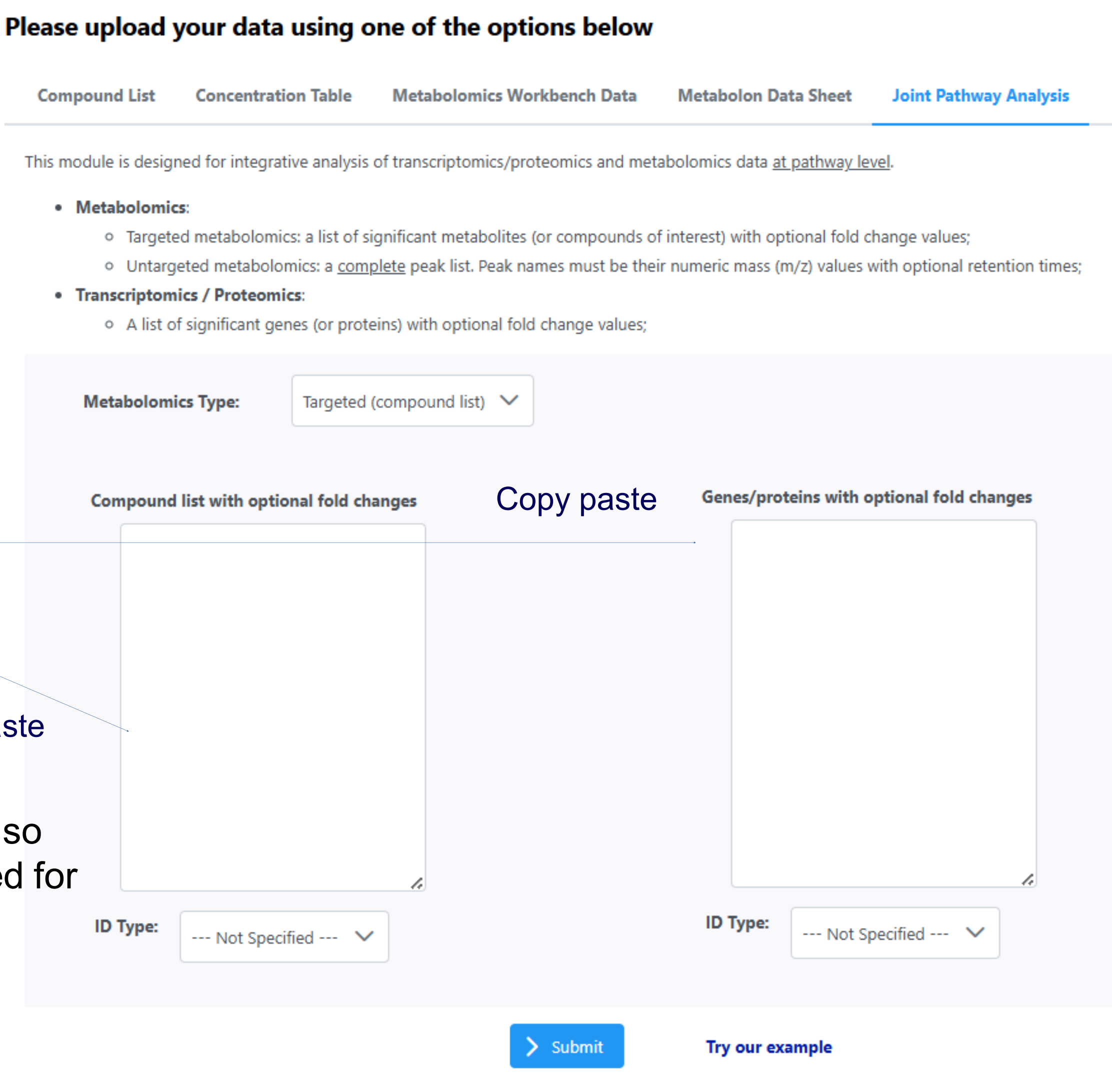


Example usage on Metaboanalyst









Copy paste

Copy paste

Multi-omics ready: when transcriptomics data are also processed in BiomiX, combined outputs are generated for Joint Pathway and Network analyses.

Please upload your data using one of the options below

Compound List | Concentration Table | Metabolomics Workbench Data | Metabolon Data Sheet | **Joint Pathway Analysis**

This module is designed for integrative analysis of transcriptomics/proteomics and metabolomics data at pathway level.

- **Metabolomics**:
  - Targeted metabolomics: a list of significant metabolites (or compounds of interest) with optional fold change values;
  - Untargeted metabolomics: a complete peak list. Peak names must be their numeric mass (m/z) values with optional retention times;
- **Transcriptomics / Proteomics**:
  - A list of significant genes (or proteins) with optional fold change values;

Metabolomics Type: Targeted (compound list)

Compound list with optional fold changes

```
HMDB0000292	-1.166874014
HMDB0000159	-0.418125595
HMDB0033951	-0.66171578
HMDB0000673_5	-0.906404543
HMDB0000925	-1.035921633
HMDB0001999	-0.82879636
HMDB0011618	-0.444430066
HMDB0000929	-0.595395106
HMDB0010379	-1.623878436
HMDB0010381	-0.429302103
HMDB0008493	-0.322414705
HMDB0008138_11	-0.656334768
HMDB0000738	-0.388022968
HMDB0000267_2	-0.30533277
HMDB0000875	-0.507805485
HMDB0001921	-0.614559907
```

ID Type: HMDB ID

Genes/proteins with optional fold changes

```
ZNF683	-2.302013132
CD69	-1.280647592
CTSW	-1.450083148
CCL4	-1.404571722
ID2	-0.719208523
FEZ1	-1.69089457
NKG7	-1.250631439
HNRNPDL	-0.607121002
FOSB	-3.053290289
PATL2	-0.709080192
G0S2	-2.305585463
CD3D	-0.933529732
NFU1	-0.63829913
XPNPEP2	-1.44088622
GZMM	-1.006252488
SPON2	-1.283466051
```

ID Type: Official Gene Symbol

Submit

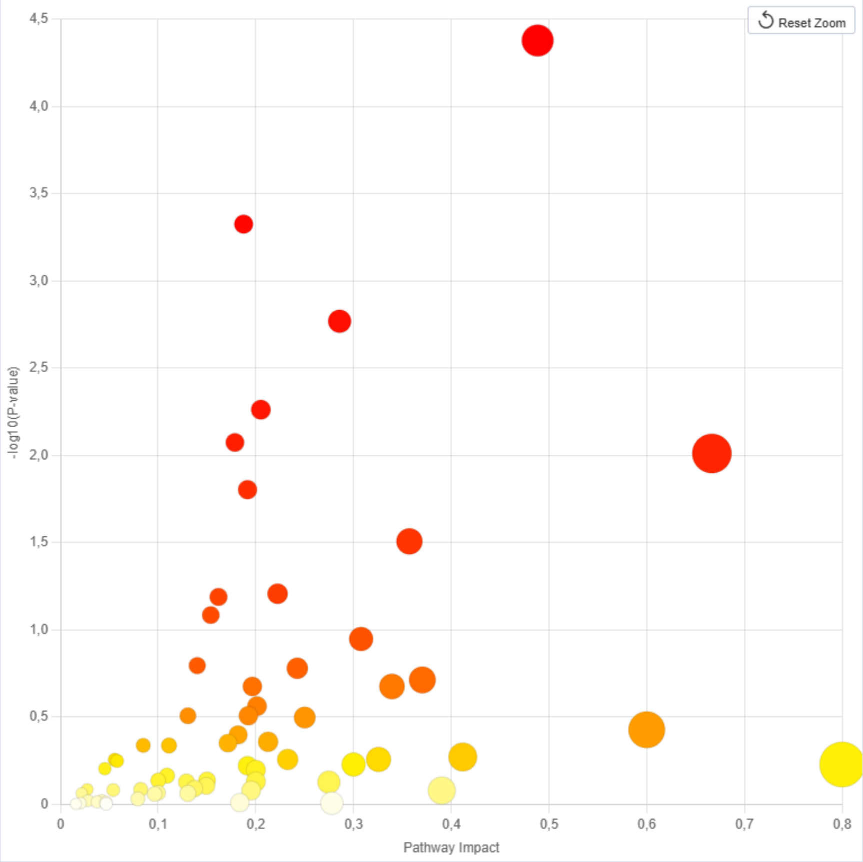
Try our example

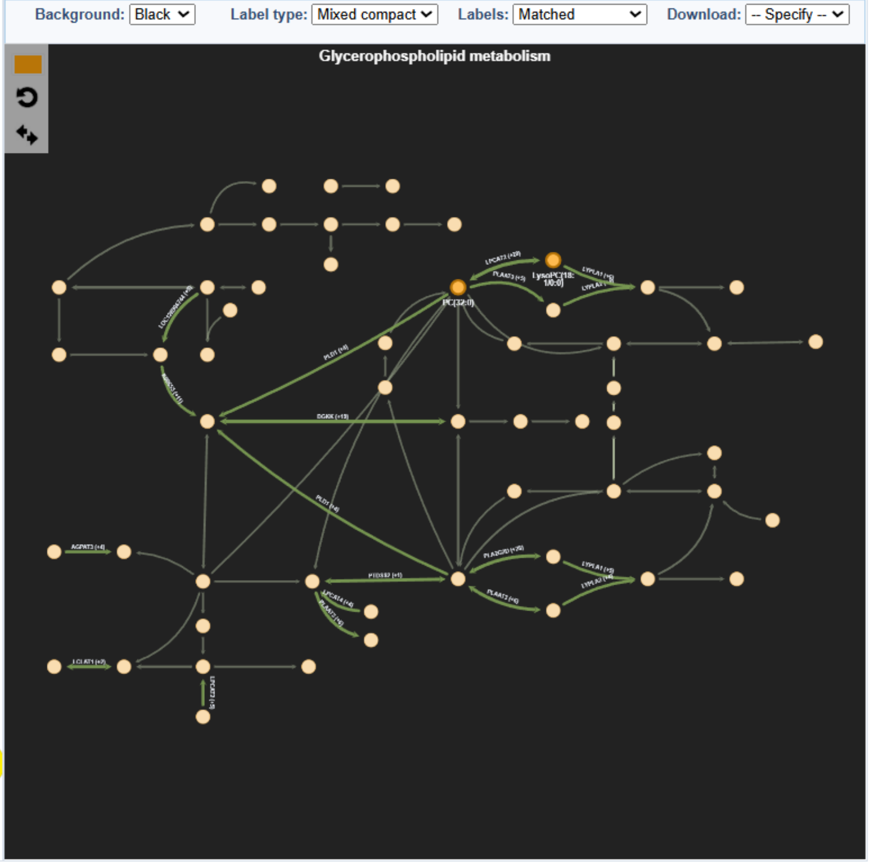
Please upload your data using one of the options below

Show Joint Pathway Analysis

Joint Pathway Analysis - Result View:

The scatter diagram summarizes matched pathways by pathway impact (x-axis) and enrichment significance (y-axis, -log10 p-value). Mouse over points to view pathway names and drag to zoom. Use reset to return to default view.





values;

tional retention times;

l fold changes

002

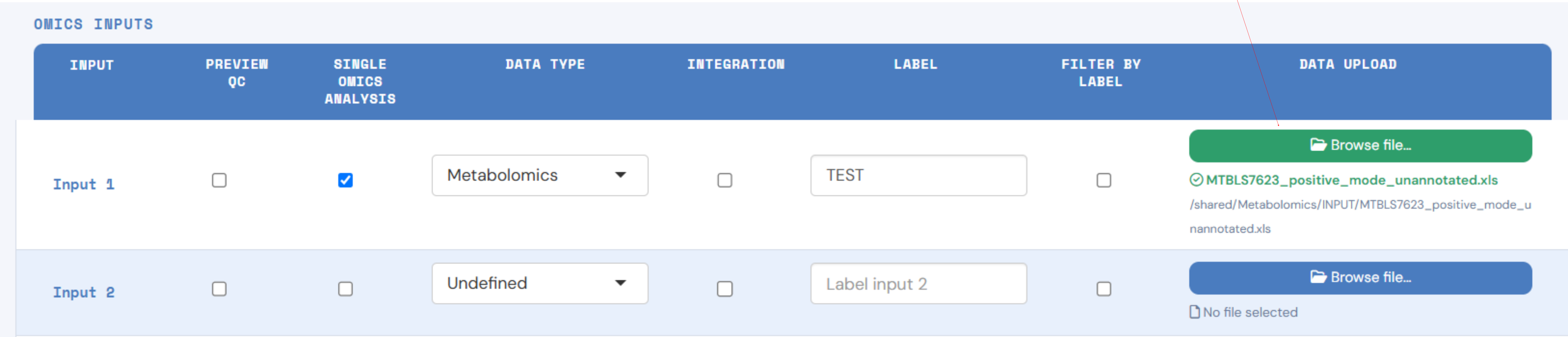
nbol

Submit

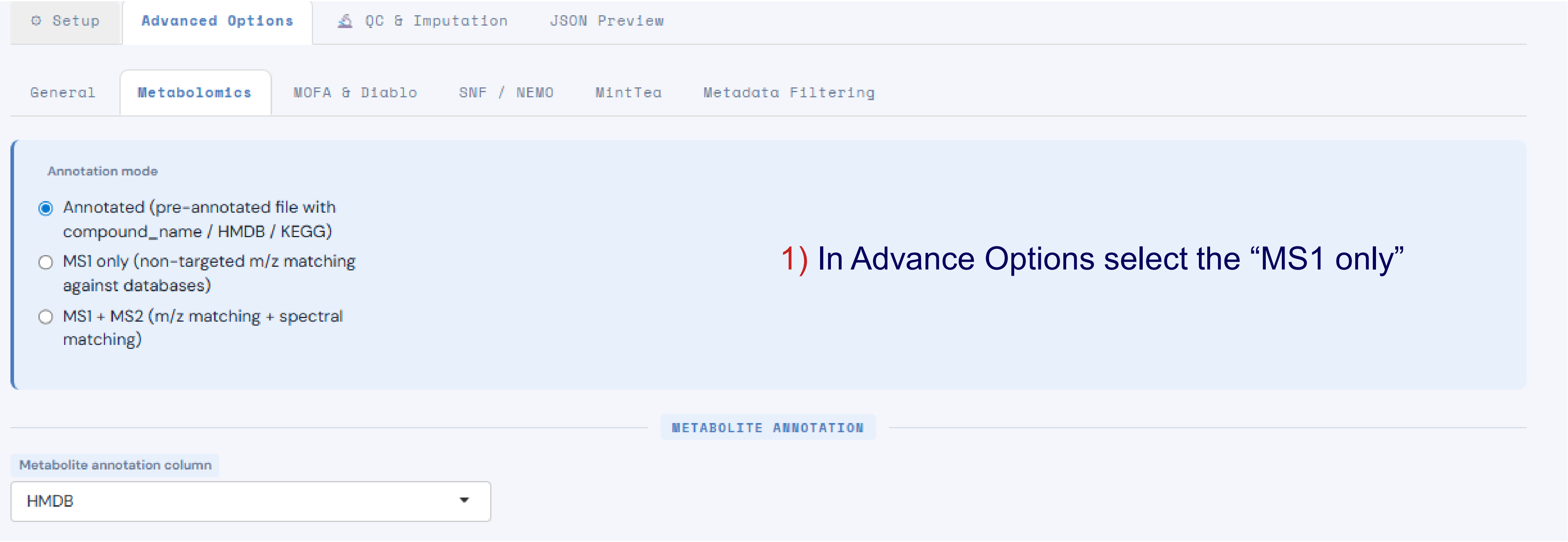
Try our example

How to annotate automatically?

2) Load MTBLS7623_positive_mode_unannotated.xls from Metabolomics folder. (Input 1)



How to annotate automatically?



1) In Advance Options select the “MS1 only”

⚙ Setup | **Advanced Options** | QC & Imputation | JSON Preview

General | **Metabolomics** | MOFA & Diablo | SNF / NEMO | MintTea | Metadata Filtering

Annotation mode

- ○ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ● MS1 only (non-targeted m/z matching against databases)
- ○ MS1 + MS2 (m/z matching + spectral matching)

Positive or Negative mode?

MS1 MATCHING PARAMETERS

Ion mode: positive

M/Z tolerance (ppm): 15

Error match tolerance between “your” peak and the database match

Which adduct do you expect?

MS1 ADDUCTS – POSITIVE MODE

- [x] M+H
- [x] M+2H
- [x] M+Na
- [x] M+K
- [x] M+NH4
- [x] M+H-H2O

MS1 ADDUCTS – NEGATIVE MODE

- [ ] M-H
- [ ] M-Cl
- [ ] M+FA-H
- [ ] M-H-H2O

Which databases you want to consult to retrive the annotation?

DATABASES MS1

- [x] HMDB
- [x] LipidMaps
- [x] Metlin
- [x] KEGG

MS1 ANNOTATION FILE (OPTIONAL)

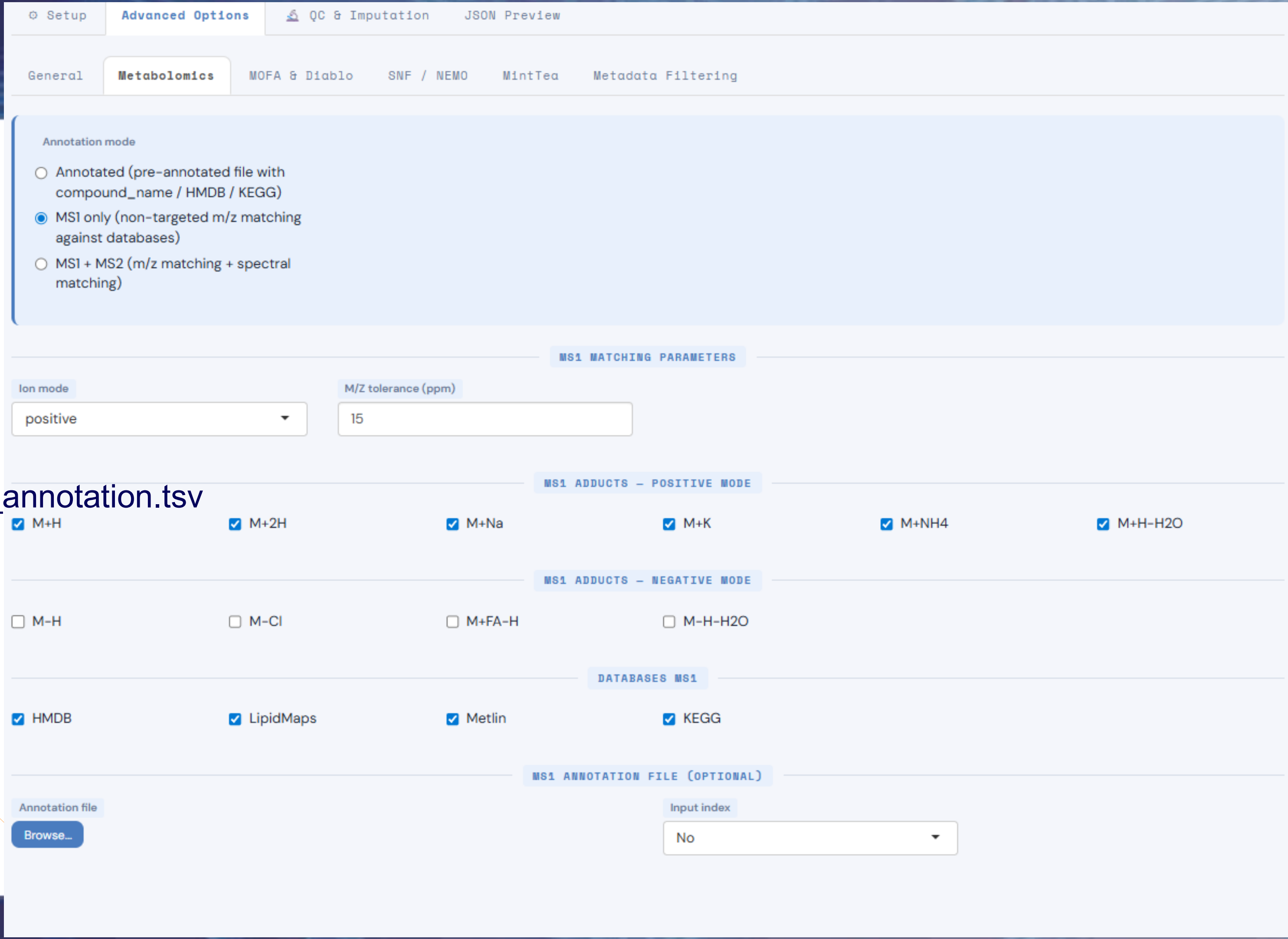
Annotation file: Browse...

The annotation should be performed on which input position?

Input index: No

3) Load MTBLS7623_positive_mode_MS1_annotation.tsv from Metabolomics folder. (Input 1)

	A	B	C
1	name	m/z	RT_min
2	peak1	70.06442	0.866483
3	peak2	76.07491	0.837358
4	peak3	86.09578	1.732158
5	peak4	86.09574	1.089042
6	peak5	91.05354	5.44075
7	peak6	104.1062	0.813258



-> Request of annotation to Ceu Mass Mediator Server

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_DOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABDOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB

+ Annotation_metabolites_B

TEST_PTB_vs_HC_results.tsv 29/06/2026 19:05 File TSV 49 KB

Adduct type

HMDB, KEGG and PubChem ID and links

Statistics

Formula

Peak number

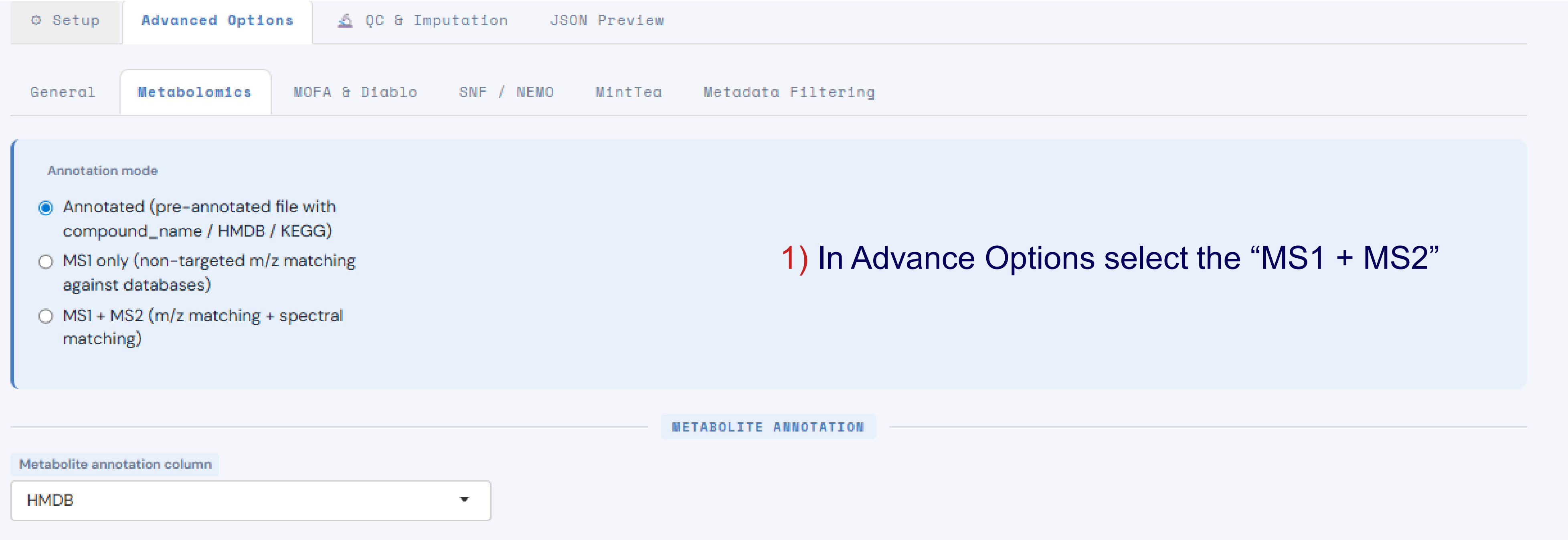
Ppm error: How much reliable?

m/z

Annotated name

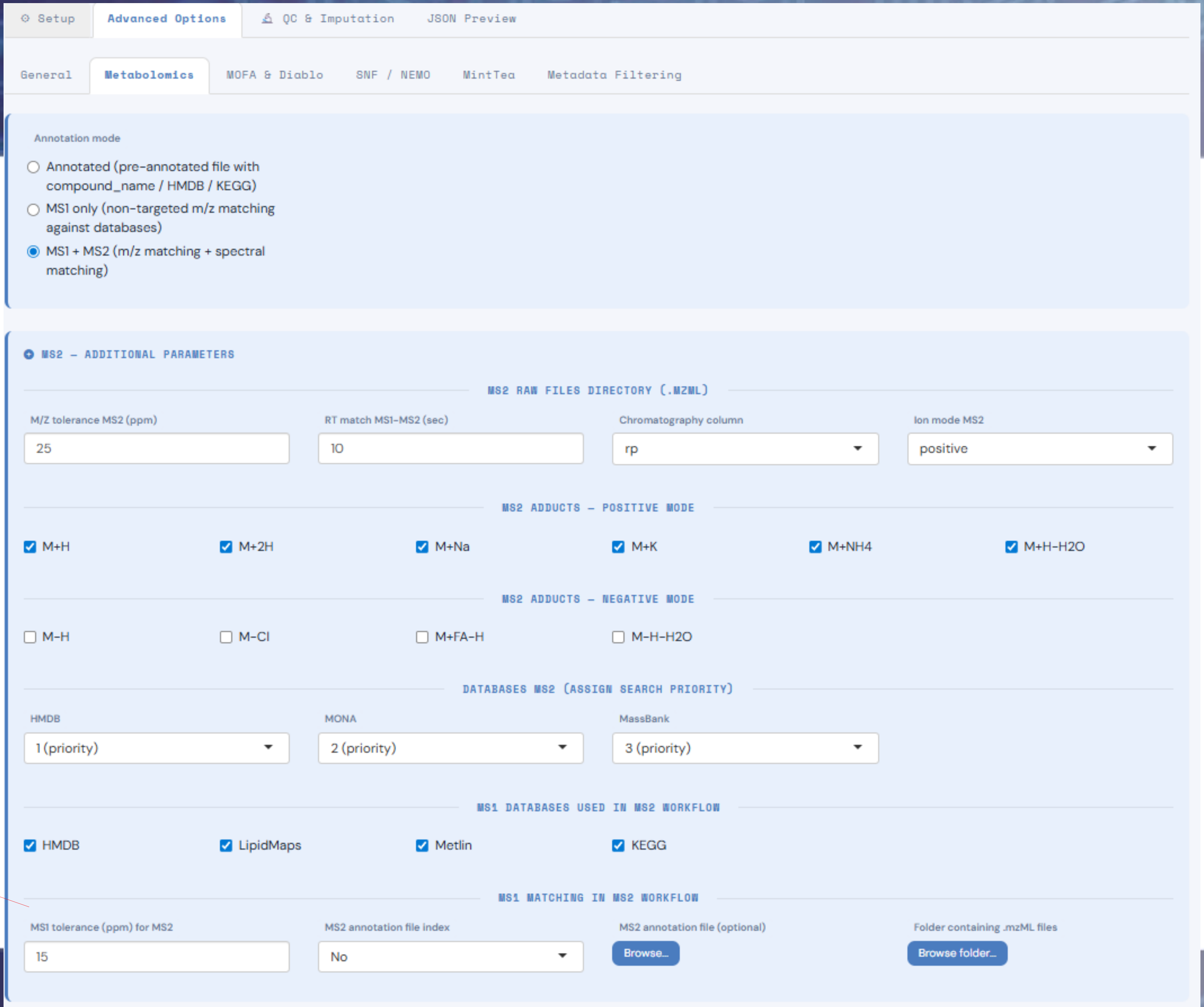
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	m/z	NAME.x	shapiro_p	p_val	log2FC	RT_sec	RT_min	padj	retention_	adduct_re	RT_score	identifier	name	formula	adduct	molecular	error_ppm	ionization	final_scor	KEGG	KEGG_UR	HMDB	HMDB_UR	LipidMaps	LipidMaps	Metlin
2	320.959	peak_114	3.614309	0.00013	#NOME?	60.265	1.00442	0.0674	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
3	452.953	peak_705	9.011311	0.00043	1.38237	69.2998	1.155	0.10778	1.155	-2	-2	117280	3-[3,5-dih	C14H10O	M+Cl	417.984	1	-2	-2			HMDB012	http://www.hmdb.ca/metabolites/HMDB			
4	452.953	peak_705	9.011311	0.00043	1.38237	69.2998	1.155	0.10778	1.155	-2	-2	58543	3-hydroxy	C14H10O	M+Cl	417.984	1	-2	-2			HMDB012	http://www.hmdb.ca/metabolites/HMDB			
5	452.953	peak_705	9.011311	0.00043	1.38237	69.2998	1.155	0.10778	1.155	-2	-2	69877	5-[3,5-dih	C14H10O	M+Cl	417.984	1	-2	-2			HMDB012	http://www.hmdb.ca/metabolites/HMDB			

How to annotate automatically using fragmentation data (MS/MS)?



1) In Advance Options select the "MS1 + MS2"

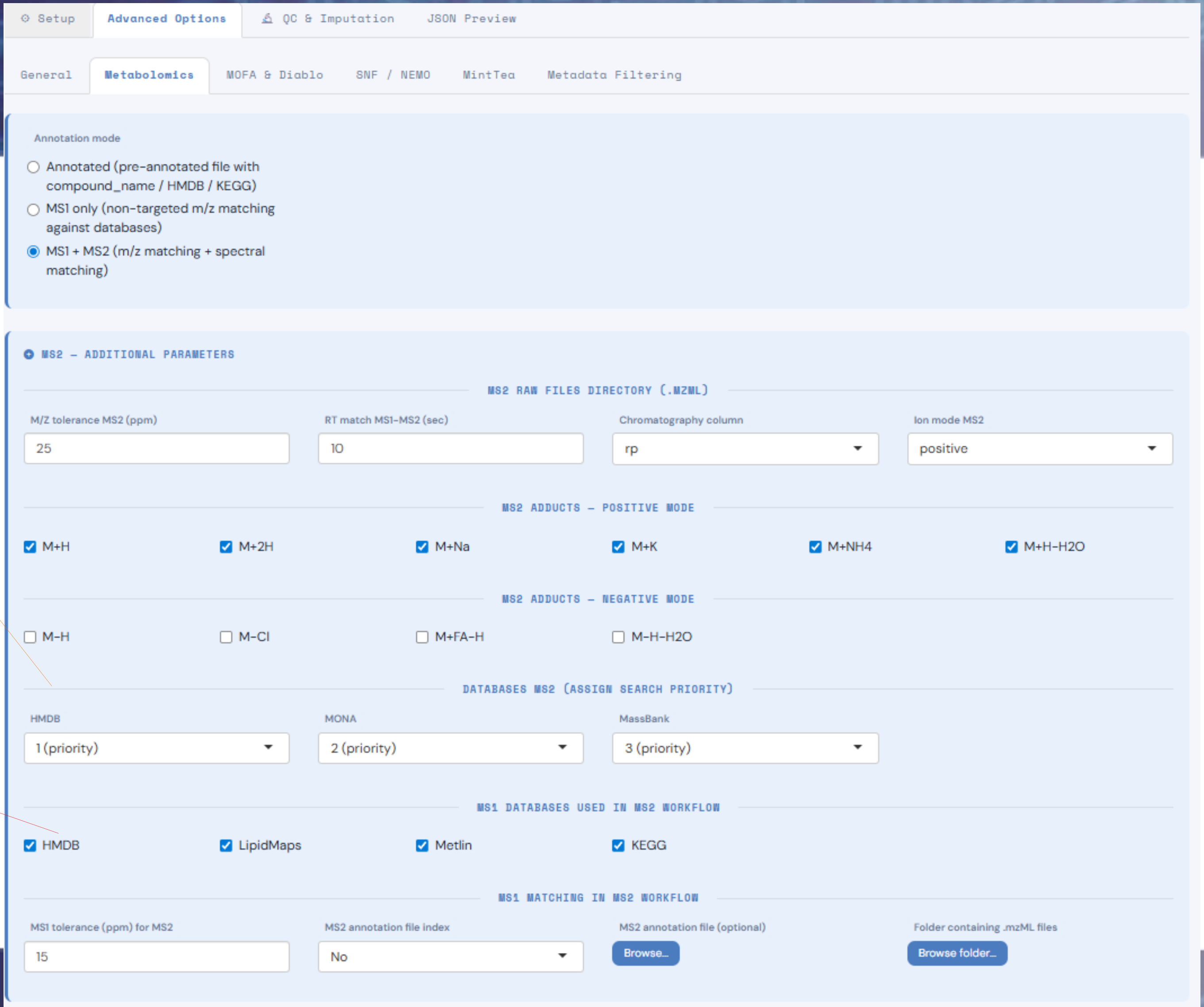
Error match tolerance between "your" peak and the database match



Error match tolerance between "your" peak m/z in MS1 and in MS2 data

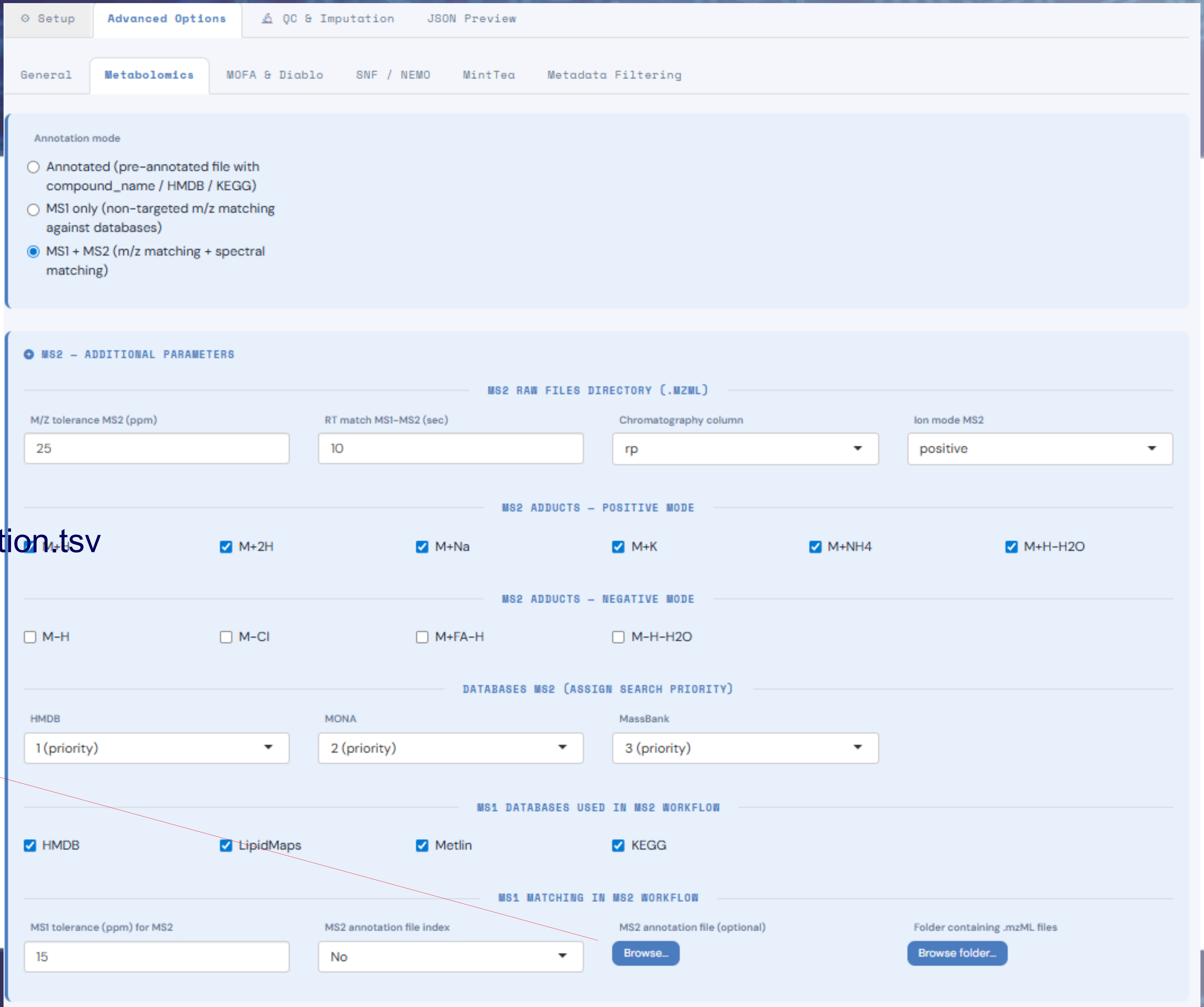
Public database priority in the MS2 match (available: HMDB, MONA, Massbank)

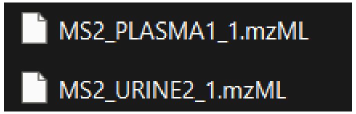
Public database priority in the MS1 match (available: HMDB, KEGG, Lipidmaps, Metlin)

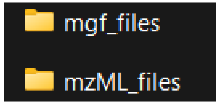


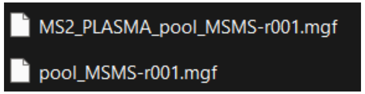
3) Load MTBLS7623_positive_mode_MS1_annotation.tsv from Metabolomics folder. (Input 1)

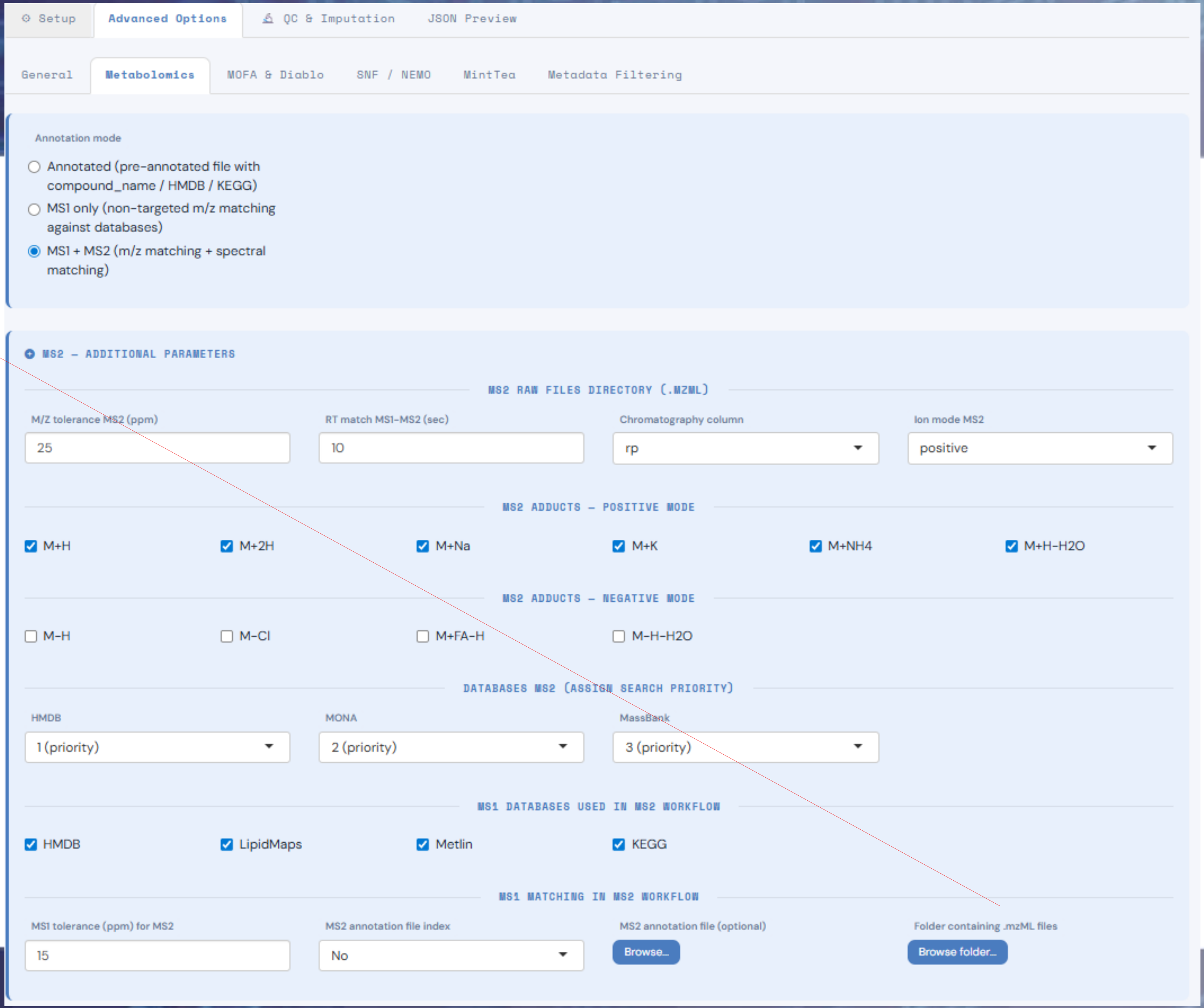
	A	B	C
1	name	m/z	RT_min
2	peak1	70.06442	0.866483
3	peak2	76.07491	0.837358
4	peak3	86.09578	1.732158
5	peak4	86.09574	1.089042
6	peak5	91.05354	5.44075
7	peak6	104.1062	0.813258











-> Provide the directory of the MzML file or mgf file(s)

Name these file(s) including the label used in the "label" i.e (TEST, or PLASMA) tidyMass

-> Request of annotation to Ceu Mass Mediator Server

-> Local MS2 spectra extraction and match on the databases selected (locally availables).

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_DOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABDOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB

+ Annotation_metabolites_B

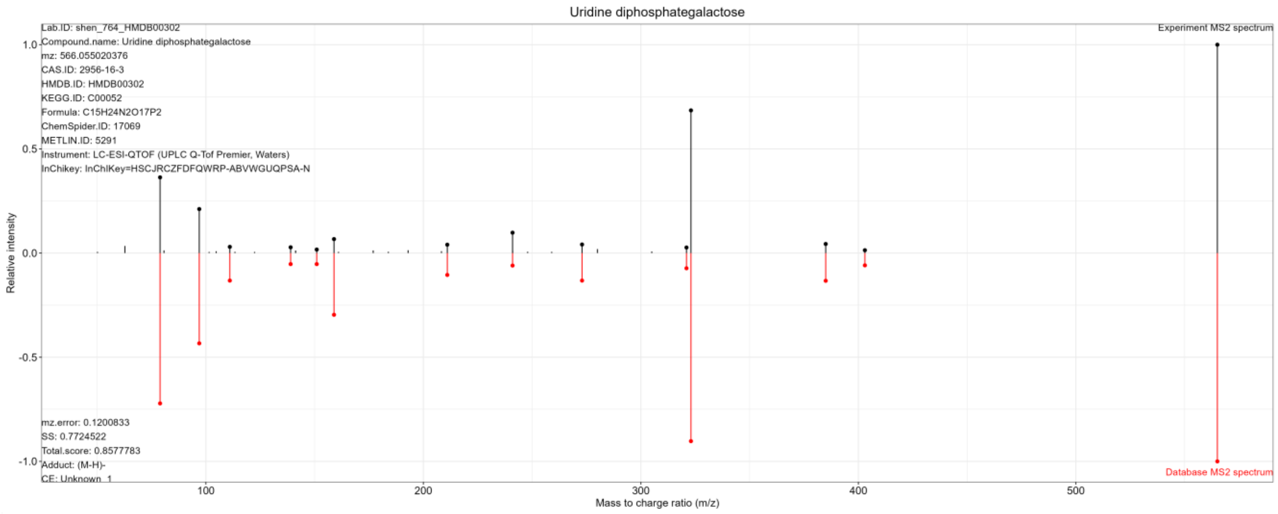
+ MS2_SPECTRA

MS2_SPECTRA

ms2_match_plot_HMDB

ms2_match_plot_MASSBANK

ms2_match_plot_MONA



HMDB_URl	LipidMaps	LipidMaps_	Metlin	Metlin_URl	PubChem	PubChem_	MS2_anno	annot_level
/dbget-bin/www_bget?cpd:C027			3596	https://me	11988267	https://pul	NA	Level_3
/dbget-bin/www_bget?cpd:C018			65671	https://me	25200469	https://pul	NA	Level_3
http://www.hmdb.ca/metabolite			62907	https://me	439795	https://pul	NA	Level_3
http://www.hmdb.ca/metabolite			3595	https://me	192	https://pul	NA	Level_3

-> Results updated with MS2 annotation and folder including spectral match of the annotated peaks

Biomix & Metabolomics - Key messages

- **Integrated & user-friendly.** QC, data analysis and visualization within a single workflow.
- **Targeted & untargeted metabolomics.** Including metabolite annotation based on MS1 and MS/MS data.
- **Interpretation-ready.** Outputs directly compatible with tools such as MetaboAnalyst.
- **Multi-omics ready.** Easy integration of metabolomics with other omics layers. (after the break)